Classification of Fusion Categories

1. Prologue

What are fusion categories? What are near-groups, Haagerup–Izumi categories and quadratic categories? What is modular data? What are $6j$ symbols? What is the even part of a subfactor? I think we need to assume that either $\mathrm{char}(\mathbb{k}) \nmid |G|$ or $\mathrm{char}(\mathbb{k}) = 0$ for the Schur orthogonality relations.

Need exposition in the deconstruction section saying that we follow Izumi's work closely using a categorical / non-unitary language.

Outline stuff: Introduction (history, why we care, overview of the paper), Preliminaries (basic definitions), Deconstruction, Irrational Reconstruction, General Reconstruction, Examples, Appendix, References

1. Introduction
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Let $\mathcal{C}$ be a skeletal fusion category over $\Bbbk$. Recall the Yoneda embedding

$$よ_*(X) \coloneqq \mathrm{Hom}_{\mathcal{C}}(-,X) \qquad \text{and} \qquad よ_*(f : X \to Y) \coloneqq (\mathrm{Hom}_{\mathcal{C}}(-,X) \Rightarrow \mathrm{Hom}_{\mathcal{C}}(-,Y)).$$

Taking the Yoneda embedding of a component $\alpha_{X,Y,Z} : (X \otimes Y) \otimes Z \to X \otimes (Y \otimes Z)$ of the associativity natural isomorphism α, we obtain a natural isomorphism

$$よ_*(\alpha_{X,Y,Z}) = (\mathrm{Hom}_{\mathcal{C}}(-,(X \otimes Y) \otimes Z) \Rightarrow \mathrm{Hom}_{\mathcal{C}}(-,X \otimes (Y \otimes Z))).$$

Thus we have an isomorphism of vector spaces

$$[よ_*(\alpha_{X,Y,Z})](W) : \mathrm{Hom}_{\mathcal{C}}(W,(X \otimes Y) \otimes Z) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}}(W, X \otimes (Y \otimes Z));$$

that is, an invertible matrix. In other words, the associativity is given by matrices indexed by X, Y, Z, W. But we can simplify things further! Suppose we now choose representatives $\{X_i\}_{i \in \Gamma}$ of isomorphism classes of simple objects and fix a choice of basis (gauge) for each *multiplicity space* $H^h_{i,j} := \mathrm{Hom}_{\mathcal{C}}(X_h, X_i \otimes X_j)$. Then by semisimplicity,

$$X_i \otimes X_j \cong \bigoplus_{h \in \Gamma} X_h^{\dim_{\Bbbk}(H^h_{i,j})},$$

whence we have both

$$\begin{aligned}\bigoplus_{m \in \Gamma} H^{i_0}_{m,i_3} \otimes_{\Bbbk} H^m_{i_1,i_2} &= \bigoplus_{m \in \Gamma} \mathrm{Hom}_{\mathcal{C}}(X_{i_0}, X_m \otimes X_{i_3}) \otimes_{\Bbbk} \mathrm{Hom}_{\mathcal{C}}(X_m, X_{i_1} \otimes X_{i_2}) \\ &\cong \mathrm{Hom}_{\mathcal{C}}(X_{i_0}, (X_{i_1} \otimes X_{i_2}) \otimes X_{i_3})\end{aligned}$$

and

$$\begin{aligned}\bigoplus_{n \in \Gamma} H^{i_0}_{i_1,n} \otimes_{\Bbbk} H^n_{i_2,i_3} &= \bigoplus_{n \in \Gamma} \mathrm{Hom}_{\mathcal{C}}(X_{i_0}, X_{i_1} \otimes X_n) \otimes_{\Bbbk} \mathrm{Hom}_{\mathcal{C}}(X_n, X_{i_2} \otimes X_{i_3}) \\ &\cong \mathrm{Hom}_{\mathcal{C}}(X_{i_0}, X_{i_1} \otimes (X_{i_2} \otimes X_{i_3})).\end{aligned}$$

We thus obtain canonical bases for the above Hom-spaces. The *(quantum) $6j$-symbols of $\mathcal{C}$* are then defined to be the matrix blocks of the form

$$\Phi^{i_0,m,n}_{i_1,i_2,i_3} : H^{i_0}_{m,i_3} \otimes_{\Bbbk} H^m_{i_1,i_2} \to H^{i_0}_{i_1,n} \otimes_{\Bbbk} H^n_{i_2,i_3}$$

such that the block-diagonal matrices $\Phi^{i_0}_{i_1,i_2,i_3} := \bigoplus_{m \in \Gamma} \bigoplus_{n \in \Gamma} \Phi^{i_0,m,n}_{i_1,i_2,i_3}$ give a change-of-basis

$$\Phi^{i_0}_{i_1,i_2,i_3} : \mathrm{Hom}_{\mathcal{C}}(X_{i_0}, (X_{i_1} \otimes X_{i_2}) \otimes X_{i_3}) \cong \mathrm{Hom}_{\mathcal{C}}(X_{i_0}, X_{i_1} \otimes (X_{i_2} \otimes X_{i_3}))$$

between the canonical bases. These matrices $\Phi^{i_0}_{i_1,i_2,i_3}$ themselves form the blocks of a block-diagonal matrix $\Phi_{i_1,i_2,i_3} := \bigoplus_{i_0 \in \Gamma} \Phi^{i_0}_{i_1,i_2,i_3}$, which gives us a change-of-basis $(X_{i_1} \otimes X_{i_2}) \otimes X_{i_3} \cong X_{i_1} \otimes (X_{i_2} \otimes X_{i_3})$ describing how the associator acts on each summand X_{i_0}. As it happens, we in fact have that

$$\Phi^{i_0,m,n}_{i_1,i_2,i_3} = v_n^{-1}(X_{i_0}, X_{i_1}, X_{i_2} \otimes X_{i_3}) \circ [よ_*(\alpha_{X_{i_1},X_{i_2},X_{i_3}})](X_{i_0}) \circ u_m(X_{i_0}, X_{i_1} \otimes X_{i_2}, X_{i_3}),$$

where

$$[u_m(W,U,V)](f \otimes_{\Bbbk} g) := (g \otimes \mathrm{id}_V) \circ f \quad \text{and} \quad [v_n(W,U,V)](f' \otimes_{\Bbbk} g') := (\mathrm{id}_U \otimes g') \circ f'$$

for all $f \in \mathrm{Hom}_{\mathcal{C}}(W, X_m \otimes V)$, $g \in \mathrm{Hom}_{\mathcal{C}}(X_m, U)$, $f' \in \mathrm{Hom}_{\mathcal{C}}(W, U \otimes X_n)$ and $g \in \mathrm{Hom}_{\mathcal{C}}(X_n, V)$).

More explicitly, suppose we have an isomorphism

$$\mathrm{Hom}_{\mathcal{C}}(X_4, (X_1 \otimes X_2) \otimes X_3) \cong \mathrm{Hom}_{\mathcal{C}}(X_4, X_1 \otimes (X_2 \otimes X_3)).$$

Let X_5 and X_6 be simple summands of $(X_1 \otimes X_2)$ and $(X_2 \otimes X_3)$, respectively. Then we can determine our isomorphism by determining the block matrices of the form

$$\mathrm{Hom}_{\mathcal{C}}(X_4, X_5 \otimes X_3) \to \mathrm{Hom}_{\mathcal{C}}(X_4, X_1 \otimes X_6)$$

for all such X_5 and X_6. These matrices are exactly the $6j$ symbols, where the six simple objects $X_1, X_2, X_3, X_4, X_5, X_6$ play the role of the eponymous "six j's". This is also where u and v come in; given $f \in H^4_{5,3}$ and $g \in H^5_{1,2}$, we have $[u_5(X_4, X_1 \otimes X_2, X_3)](f \otimes_{\Bbbk} g) : X_4 \to (X_1 \otimes X_2) \otimes X_3$, and similarly $[v_6(X_4, X_1, X_2 \otimes X_3)](f' \otimes_{\Bbbk} g') : X_4 \to X_1 \otimes (X_2 \otimes X_3)$ for $f' \in H^4_{1,6}$ and $g' \in H^6_{2,3}$.

Example 1.1. ($\mathsf{Vec}_{\mathbb{Z}/2\mathbb{Z}}$). Consider $\mathsf{Vec}_{\mathbb{Z}/2\mathbb{Z}}$, a category with two isomorphism classes of simple objects – $[\mathbb{1}]$ and $[X]$ – satisfying the fusion rule

$$[X] \otimes [X] = [\mathbb{1}].$$

Let's assume that it's skeletal – that is, $X \otimes X = \mathbb{1}$ – and start by computing the change-of-basis matrix $\Phi_{X,X,X}$. This matrix can be written in block diagonal form as

$$\Phi_{X,X,X} = \begin{pmatrix} \Phi^{\mathbb{1}}_{X,X,X} & 0 \\ 0 & \Phi^{X}_{X,X,X} \end{pmatrix}.$$

First, observe that

$$\Phi^{\mathbb{1}}_{X,X,X} : \mathrm{Hom}(\mathbb{1}, (X \otimes X) \otimes X) \to \mathrm{Hom}(\mathbb{1}, X \otimes (X \otimes X));$$

but $(X \otimes X) \otimes X = X = X \otimes (X \otimes X)$, so $\Phi^{\mathbb{1}}_{X,X,X}$ is just the isomorphism of 0-dimensional vector spaces; that is, $\Phi^{\mathbb{1}}_{X,X,X} = 0$. Let's now compute $\Phi^{X}_{X,X,X}$. We shall start by choosing bases for the multiplicity spaces $H^{\mathbb{1}}_{X,X}$, $H^{X}_{\mathbb{1},X}$ and $H^{X}_{X,\mathbb{1}}$. These spaces are all 1-dimensional, so we will choose basis elements $\iota^{\mathbb{1}}_{X,X}$, λ^{-1}_X and ρ^{-1}_X, respectively. Here we have a canonical choice of basis elements for $H^{X}_{\mathbb{1},X}$ and $H^{X}_{X,\mathbb{1}}$; namely, the left and right unitors λ and ρ. This culminates in the basis elements

$$\iota^{X}_{(X\otimes X)\otimes X} \coloneqq (\iota^{\mathbb{1}}_{X,X} \otimes \mathrm{id}_X) \circ \lambda^{-1}_X = [u_{\mathbb{1}}(X, X \otimes X, X)](\lambda^{-1}_X \otimes_{\Bbbk} \iota^{\mathbb{1}}_{X,X}),$$

$$\iota^{X}_{X\otimes(X\otimes X)} \coloneqq (\mathrm{id}_X \otimes \iota^{\mathbb{1}}_{X,X}) \circ \rho^{-1}_X = [v_{\mathbb{1}}(X, X, X \otimes X)](\rho^{-1}_X \otimes_{\Bbbk} \iota^{\mathbb{1}}_{X,X})$$

for $\mathrm{Hom}(X, (X \otimes X) \otimes X)$ and $\mathrm{Hom}(X, X \otimes (X \otimes X))$, respectively. These are once again both 1-dimensional spaces, so let's define a constant $\omega_1 \in \Bbbk^\times$ for which

$$\Phi^{X}_{X,X,X}(\iota^{X}_{(X\otimes X)\otimes X}) \eqqcolon \omega_1 \cdot \iota^{X}_{X\otimes(X\otimes X)}.$$

To gain constraints on ω_1, we must use the pentagon equation. To this end, consider the Hom-spaces

$$\mathrm{Hom}(\mathbb{1}, (\mathbb{1} \otimes X) \otimes X),$$
$$\mathrm{Hom}(\mathbb{1}, \mathbb{1} \otimes (X \otimes X)),$$
$$\mathrm{Hom}(\mathbb{1}, (X \otimes \mathbb{1}) \otimes X),$$
$$\mathrm{Hom}(\mathbb{1}, X \otimes (\mathbb{1} \otimes X)),$$
$$\mathrm{Hom}(\mathbb{1}, (X \otimes X) \otimes \mathbb{1}),$$
$$\mathrm{Hom}(\mathbb{1}, X \otimes (X \otimes \mathbb{1})).$$

We can write bases for these Hom-spaces in terms of the bases we chose previously. That is,

$$\begin{aligned}
\iota^{\mathbb{1}}_{(\mathbb{1}\otimes X)\otimes X} &\coloneqq (\lambda_X^{-1}\otimes \mathrm{id}_X)\circ \iota^{\mathbb{1}}_{X,X},\\
\iota^{\mathbb{1}}_{\mathbb{1}\otimes (X\otimes X)} &\coloneqq (\mathrm{id}_{\mathbb{1}}\otimes \iota^{\mathbb{1}}_{X,X})\circ \lambda_{\mathbb{1}}^{-1},\\
\iota^{\mathbb{1}}_{(X\otimes \mathbb{1})\otimes X} &\coloneqq (\rho_X^{-1}\otimes \mathrm{id}_X)\circ \iota^{\mathbb{1}}_{X,X},\\
\iota^{\mathbb{1}}_{X\otimes (\mathbb{1}\otimes X)} &\coloneqq (\mathrm{id}_X\otimes \lambda_X^{-1})\circ \iota^{\mathbb{1}}_{X,X},\\
\iota^{\mathbb{1}}_{(X\otimes X)\otimes \mathbb{1}} &\coloneqq (\iota^{\mathbb{1}}_{X,X}\otimes \mathrm{id}_{\mathbb{1}})\circ \lambda_{\mathbb{1}}^{-1},\\
\iota^{\mathbb{1}}_{X\otimes (X\otimes \mathbb{1})} &\coloneqq (\mathrm{id}_X\otimes \rho_X^{-1})\circ \iota^{\mathbb{1}}_{X,X}.
\end{aligned}$$

This culminates in the three new constants given by

$$\begin{aligned}
\Phi^{\mathbb{1}}_{\mathbb{1},X,X}(\iota^{\mathbb{1}}_{(\mathbb{1}\otimes X)\otimes X}) &\eqqcolon \omega_2\cdot \iota^{\mathbb{1}}_{\mathbb{1}\otimes (X\otimes X)},\\
\Phi^{\mathbb{1}}_{X,\mathbb{1},X}(\iota^{\mathbb{1}}_{(X\otimes \mathbb{1})\otimes X}) &\eqqcolon \omega_3\cdot \iota^{\mathbb{1}}_{X\otimes (\mathbb{1}\otimes X)},\\
\Phi^{\mathbb{1}}_{X,X,\mathbb{1}}(\iota^{\mathbb{1}}_{(X\otimes X)\otimes \mathbb{1}}) &\eqqcolon \omega_4\cdot \iota^{\mathbb{1}}_{X\otimes (X\otimes \mathbb{1})}.
\end{aligned}$$

The final bases we need are for the Hom-spaces

$$\begin{gathered}
\mathrm{Hom}(\mathbb{1}, ((X\otimes X)\otimes X)\otimes X),\\
\mathrm{Hom}(\mathbb{1}, (X\otimes (X\otimes X))\otimes X),\\
\mathrm{Hom}(\mathbb{1}, (X\otimes X)\otimes (X\otimes X)),\\
\mathrm{Hom}(\mathbb{1}, X\otimes ((X\otimes X)\otimes X)),\\
\mathrm{Hom}(\mathbb{1}, X\otimes (X\otimes (X\otimes X))).
\end{gathered}$$

Following the same procedure as before, we have

$$\begin{aligned}
\iota^{\mathbb{1}}_{((X\otimes X)\otimes X)\otimes X} &\coloneqq (\iota^{X}_{(X\otimes X)\otimes X}\otimes \mathrm{id}_X)\circ \iota^{\mathbb{1}}_{X,X} = ((\iota^{\mathbb{1}}_{X,X}\otimes \mathrm{id}_X)\otimes \mathrm{id}_X)\circ \iota^{\mathbb{1}}_{(\mathbb{1}\otimes X)\otimes X},\\
\iota^{\mathbb{1}}_{(X\otimes (X\otimes X))\otimes X} &\coloneqq (\iota^{X}_{X\otimes (X\otimes X)}\otimes \mathrm{id}_X)\circ \iota^{\mathbb{1}}_{X,X} = ((\mathrm{id}_X\otimes \iota^{\mathbb{1}}_{X,X})\otimes \mathrm{id}_X)\circ \iota^{\mathbb{1}}_{(X\otimes \mathbb{1})\otimes X},\\
\iota^{\mathbb{1}}_{(X\otimes X)\otimes (X\otimes X)} &\coloneqq (\iota^{\mathbb{1}}_{X,X}\otimes (\mathrm{id}_X\otimes \mathrm{id}_X))\circ \iota^{\mathbb{1}}_{\mathbb{1}\otimes (X\otimes X)} = ((\mathrm{id}_X\otimes \mathrm{id}_X)\otimes \iota^{\mathbb{1}}_{X,X})\circ \iota^{\mathbb{1}}_{(X\otimes X)\otimes \mathbb{1}},\\
\iota^{\mathbb{1}}_{X\otimes ((X\otimes X)\otimes X)} &\coloneqq (\mathrm{id}_X\otimes \iota^{X}_{(X\otimes X)\otimes X})\circ \iota^{\mathbb{1}}_{X,X} = (\mathrm{id}_X\otimes (\iota^{\mathbb{1}}_{X,X}\otimes \mathrm{id}_X))\circ \iota^{\mathbb{1}}_{X\otimes (\mathbb{1}\otimes X)},\\
\iota^{\mathbb{1}}_{X\otimes (X\otimes (X\otimes X))} &\coloneqq (\mathrm{id}_X\otimes \iota^{X}_{X\otimes (X\otimes X)})\circ \iota^{\mathbb{1}}_{X,X} = (\mathrm{id}_X\otimes (\mathrm{id}_X\otimes \iota^{\mathbb{1}}_{X,X}))\circ \iota^{\mathbb{1}}_{X\otimes (X\otimes \mathbb{1})}.
\end{aligned}$$

The pentagon diagram thus gives us

$$\begin{tikzcd}
 & & \iota^{\mathbb{1}}_{((X\otimes X)\otimes X)\otimes X} \arrow[dll, mapsto, "\alpha_{X,X,X}\otimes \mathrm{id}_X"'] \arrow[drr, mapsto, "\alpha_{X\otimes X,X,X}"] & & \\
\omega_1\cdot \iota^{\mathbb{1}}_{(X\otimes (X\otimes X))\otimes X} \arrow[d, mapsto, "\alpha_{X,X\otimes X,X}"'] & & & & \omega_2\cdot \iota^{\mathbb{1}}_{(X\otimes X)\otimes (X\otimes X)} \arrow[d, mapsto, "\alpha_{X,X,X\otimes X}"] \\
\omega_1\omega_3\cdot \iota^{\mathbb{1}}_{X\otimes ((X\otimes X)\otimes X)} \arrow[rr, mapsto, "\mathrm{id}_X\otimes \alpha_{X,X,X}"'] & & \omega_1^2\omega_3\cdot \iota^{\mathbb{1}}_{X\otimes (X\otimes (X\otimes X))} \arrow[rr, equal] & & \omega_2\omega_4\cdot \iota^{\mathbb{1}}_{X\otimes (X\otimes (X\otimes X))}
\end{tikzcd}.$$

This gives us the relation $\omega_1^2\omega_3 = \omega_2\omega_4$. However, there could be more constraints on these constants! As it turns out, though, there is a simple way of determining these. Suppose we use the more insightful labeling convention

$$\begin{aligned}\omega(X,X,X) &\coloneqq \omega_1,\\ \omega(\mathbb{1},X,X) &\coloneqq \omega_2,\\ \omega(X,\mathbb{1},X) &\coloneqq \omega_3,\\ \omega(X,X,\mathbb{1}) &\coloneqq \omega_4.\end{aligned}$$

The reason why we have chosen this relabeling is because now acting by $\alpha_{A,B,C}$ in the pentagon diagram introduces a factor of $\omega(A,B,C)$. If we apply this to the general pentagon diagram

$$\begin{tikzcd}
 & ((A\otimes B)\otimes C)\otimes D \arrow[dl, "\alpha_{A,B,C}\otimes \mathrm{id}_D"'] \arrow[dr, "\alpha_{A\otimes B,C,D}"] & \\
(A\otimes(B\otimes C))\otimes D \arrow[d, "\alpha_{A,B\otimes C,D}"'] & & (A\otimes B)\otimes(C\otimes D) \arrow[d, "\alpha_{A,B,C\otimes D}"] \\
A\otimes((B\otimes C)\otimes D) \arrow[rr, "\mathrm{id}_A\otimes\alpha_{B,C,D}"'] & & A\otimes(B\otimes(C\otimes D))
\end{tikzcd},$$

we obtain the constraints

$$\omega(A,B,C)\omega(A,BC,D)\omega(B,C,D) = \omega(AB,C,D)\omega(A,B,CD)$$

for all $A,B,C,D \in \mathrm{Ob}(\mathsf{Vec}_{\mathbb{Z}/2\mathbb{Z}})$. This is exactly the definition for a 3-cocycle $\omega : (\mathbb{Z}/2\mathbb{Z})^{\oplus 3} \to \Bbbk^\times$! In general, given a finite Abelian group G, there is a bijective correspondence between cohomologous 3-cocycles on G and monoidal equivalence classes of $6j$-symbols. We shall henceforth denote by Vec_G^ω the category of G-graded vector spaces whose $6j$-symbols are given by the non-trivial 3-cocycle $\omega : G\times G\times G \to \Bbbk^\times$, and Vec_G the category whose $6j$-symbols are trivial.

Consider Vec_G^ω, for some non-trivial 3-cocycle ω. By the strictification theorem of Mac Lane, this category is monoidally equivalent to a strict monoidal category, say $\mathcal{S}$. But if the associator for $\mathcal{S}$ is trivial, wouldn't the $6j$-symbols too be trivial? Well, suppose that for any pair $g,h\in G$, we pick an isomorphism $\mu_{g,h} : g\otimes h \to gh$. Note that in the skeletal setting $g\otimes h = gh$, but in the strict setting they are only guaranteed to be isomorphic. Because $\mathrm{Hom}(g\otimes(h\otimes k),(g\otimes h)\otimes k)$ is 1-dimensional,

$$\mu_{g,hk}\circ(\mathrm{id}_g\otimes\mu_{h,k}) = \omega(g,h,k)\mu_{gh,k}\circ(\mu_{g,h}\otimes\mathrm{id}_k)$$

for some $\omega(g,h,k)\in\Bbbk^\times$. Doing this for each vertex of the pentagon diagram, we recover that ω must be a 3-cocycle. If we choose another isomorphism, say $\mu'_{g,h}$, then it must satisfy $\mu'_{g,h} = \beta(g,h)\mu_{g,h}$ for some $\beta(g,h)\in\Bbbk^\times$. Replacing μ with μ' gives us a new scalar

$$\omega'(g,h,k) = \beta(h,k)^{-1}\beta(g,hk)^{-1}\beta(gh,k)\beta(g,h)\omega(g,h,k).$$

In other words, it multiplies ω by a 3-coboundary, giving us a cohomologous 3-cocycle.

2. The Cuntz Algebra Approach of Izumi

Consider the category $\mathrm{End}(M)$, for M a hyperfinite type III factor. This category is strict, as $\rho \otimes \sigma \coloneqq \rho \circ \sigma$ by definition. Every near-group category with group G contains some copy of Vec_G^ω corresponding to the group-like part. Because every unitary near-group category is a subcategory of $\mathrm{End}(M)$ and is hence itself strict, we know that it will actually contain the "strictification" of some Vec_G^ω. However, Izumi shows that if $\mathcal{C}$ is any fusion category containing a simple object that is fixed under tensor products with invertibles (that is, there exists some simple object X such that $X \otimes g \cong X$ for all invertible g), then it contains a copy of Vec_G, for G the group of isomorphism classes of invertible objects. He shows in addition that if the fusion category is also unitary, then $g \otimes X = X$ (but we may not necessarily have that $X \otimes g = X$). The upshot is that we almost know how objects are tensored, since the group-like part will have trivial associativity (that is, $g \otimes h = gh$). We just need to understand $X \otimes g$ and $X \otimes X$, as well as the morphisms.

In [Izu17], Izumi showed that every unitary near-group category $\mathcal{C}$ with multiplicity m is equivalent to a subcategory of $\mathrm{End}(M)$, where M is the hyperfinite type III_1 factor. In particular, it is generated by a single irreducible endomorphism $\rho \in \mathrm{End}_0(M)$ satisfying the fusion rules

$$\begin{aligned}
[\alpha_g] \otimes [\alpha_h] &= [\alpha_{gh}], \\
[\alpha_g] \otimes [\rho] = [\rho] \otimes [\alpha_g] &= [\rho], \\
[\rho] \otimes [\rho] &= \bigoplus_{g \in G} [\alpha_g] \oplus [\rho]^{\oplus m},
\end{aligned}$$

where the map $\alpha : G \to \mathrm{Aut}(M)$ induces an injective homomorphism from G into $\mathrm{Out}(M)$.

The main result of [Izu17] is [Izu17, Theorem 4.9]. Essentially, there is a bijective correspondence between the set of equivalence classes of unitary near-group categories with finite group G and multiplicity parameter m and the set of equivalence classes of admissible tuples $(\mathcal{K}, j_1, j_2, V, U_{\mathcal{K}}, \chi, l)$ (see [Izu17, Definition 4.8]). Here $\mathcal{K}$ is the finite-dimensional Hilbert space $\mathrm{Hom}(\rho, \rho^2)$, j_1 and j_2 are two antilinear isometries of $\mathcal{K}$, V and $U_{\mathcal{K}}$ are unitary representations of G on $\mathcal{K}$, $\{\chi_g\}_{g \in G}$ are characters of G and l is a linear map from $\mathcal{K}$ to the set $\mathcal{B}(\mathcal{K}, \mathcal{K} \otimes \mathcal{K})$ of bounded operators $\mathcal{K} \to \mathcal{K} \otimes \mathcal{K}$.

By [Izu17, Theorem 9.1], the unitary near-group categories with finite Abelian group G and $m = |G|$ are completely classified tuples of the form $(\langle \cdot, \cdot \rangle, a, b, c)$, where $\langle \cdot, \cdot \rangle : G \times G \to \mathbb{T}$ is a non-degenerate symmetric bicharacter and where $a : G \to \mathbb{T}$, $b : G \to \mathbb{T}$ and $c \in \mathbb{T}$ satisfy various conditions. When we say that $\langle \cdot, \cdot \rangle$ is a bicharacter, we mean that

$$\langle xy, z \rangle = \langle x, z \rangle \langle y, z \rangle \qquad \text{and} \qquad \langle x, yz \rangle = \langle x, y \rangle \langle x, z \rangle$$

for all $x, y, z \in G$. By non-degenerate, we mean that

$$\langle x, \cdot \rangle = \langle y, \cdot \rangle$$

if and only if $x = y$. This is equivalent to the map $\varphi : G \to \mathrm{Hom}(G, \mathbb{T})$ given by $x \mapsto \langle x, \cdot \rangle$ being an isomorphism.

Definition 2.1. (Cuntz Algebra). *Let* $\{S_i\}_{i=1}^n$ *be a set of isometries on an infinite-dimensional Hilbert space* $\mathscr{H}$*. Suppose moreover that these isometries satisfy the* Cuntz relation

$$\sum_{k=1}^{n} S_k S_k^* = 1.$$

The Cuntz algebra $\mathcal{O}_n$ *is the universal* C^**-algebra* $C^*(S_1, \ldots, S_n)$*.*

Remark 2.2. Note that, as isometries, $S_i^* S_i = 1$. In particular, we must have that $S_i^* S_j = \delta_{i,j}$ for all $i, j \in \{1, \ldots, n\}$. This follows from the fact that a sum of projections is itself a projection if and only if the projections in the sum are pairwise orthogonal. The Cuntz relation is essentially ensuring that the sum of the projections $S_i S_i^*$ is the trivial projection.

Example 2.3. (Fibonacci Category). Let's look at the Fibonacci category. This is the near-group with $G = \{0\}$ and $m = 1$. Our choice for $\langle \cdot, \cdot \rangle$ is obvious, and [Izu17, Lemma 7.1] tells us that

$$c^3 a(0) = \sqrt{n} = 1 \implies a(0) = c^{-3}.$$

Moreover, [Izu17, Theorem 9.1] tells us that b is defined by $b : 0 \mapsto -1/d$, where d corresponds to the dimension of our irreducible generator ρ. Let's determine c and d. Because b is equal to its own Fourier transform, [Izu17, Theorem 9.1] tells us that

$$b(0) = ca(0)b(0) \implies a(0) = c^{-1}.$$

In order for $c^{-1} = c^{-3}$, we require $c = \pm 1$. Finally, [Izu17, Equation (9.5)] tells us that

$$\begin{aligned} b(0)b(0)b(0) &= b(0)b(0) \mp \frac{1}{d}, \\ \implies -\frac{1}{d^3} &= \frac{1}{d^2} \mp \frac{1}{d}, \\ \implies \pm d^2 - d - 1 &= 0. \end{aligned}$$

This only has a real solution when $c = 1$, whence d is nothing but the golden ratio (as it cannot be negative in the unitary case - this alternative solution, known as the Galois dual, corresponds to a non-unitary near-group in this case and many others). This is exactly what we would expect, as d is the dimension of X (where $d^2 = 1 + d$ comes from the fusion rule $X^2 = \mathbb{1} \oplus X$).

Example 2.4. ($G = \mathbb{Z}/2\mathbb{Z}$). Let's look at the case where $G = \mathbb{Z}/2\mathbb{Z}$ and $m = 2$. This near-group corresponds to the even part of the type A_4 subfactor. We know the dimension is

$$d_\pm := \frac{m \pm \sqrt{m^2 + 4n}}{2} = 1 \pm \sqrt{3}.$$

In the unitary setting, we of course ask that d be positive, and hence we choose $d = d_+$. The only possibility for a non-degenerate bicharacter is

$$\langle 0, 0 \rangle = 1, \qquad \langle 0, 1 \rangle = \langle 1, 0 \rangle = 1 \qquad \text{and} \qquad \langle 1, 1 \rangle = -1.$$

From [Izu17, Equation (7.8)], it follows that

$$a(0) = 1 \qquad \text{and} \qquad a(1) = \pm i.$$

Meanwhile, [Izu17, Equation (9.4)] tells us that

$$\overline{b(1)} = \pm i b(1) \implies \Re(b(1)) = \mp \Im(b(1)),$$

whence [Izu17, Equation (9.3)] gives us

$$\Re(b(1))^2 + \Im(b(1))^2 = (b(1)\overline{b(1)})^2 = \frac{1}{2} \implies b(1) = \frac{1 - a(1)}{2}.$$

It then follows from evaluating [Izu17, Equation (9.1)] with $g = 0$ and rearranging for c that

$$c = \frac{1 - \sqrt{3} + a(1)(1 + \sqrt{3})}{2\sqrt{2}}.$$

Note that we may choose either $a(1) = i$ or $a(1) = -i$; both of these lead to solutions. Moreover, in the non-unitary setting, we may take the Galois conjugate of d.

Example 2.5. ($G = \mathbb{Z}/2\mathbb{Z}$). Let's determine the Haagerup–Izumi categories with $G = \mathbb{Z}/2\mathbb{Z}$. Let

$$d_\pm := \frac{n \pm \sqrt{n^2+4}}{2},$$

where in this example $d := 1+\sqrt{2}$. Izumi's classification involves a triplet $(\epsilon_h(g), \omega(g), A_{h,k}(g))$, where $\epsilon_h(g) \in \{-1,1\}$, $\omega(g) \in \mathbb{T}$ and $A_{h,k}(g) \in \mathbb{C}$ satisfy [Izu18, Equations (4.1)–(4.9)]. Well, we know

$$\epsilon_0(0) = \epsilon_1(0) = 1 \qquad \text{and} \qquad \epsilon_0(1) = \epsilon_0(1)\epsilon_0(1) \implies \epsilon_0(1) = 1.$$

By [Izu18, Equation (4.7)],

$$A_{0,0}(g) = A_{0,0}(g)\omega(g),$$

which tells us that either $\omega(g) = 1$ or $A_{0,0}(g) = 0$ for each $g \in G$. Let's fix any $g \in G$ and consider the case when $A_{0,0}(g) = 0$. In this case, however, [Izu18, Equations (4.3) and (4.4)] give us

$$A_{1,0}(g)\overline{A_{\delta_{g,0}-g,0}(g)} = 1 - \frac{|\omega(g)|}{d} \implies \text{“}\frac{1}{d^2} = 1 - \frac{1}{d}\text{”}.$$

This “equality” is nonsense; we must therefore have $\omega(g) = 1$ for all $g \in G$. Suppose now that $\epsilon_1(1) = 1$. Then [Izu18, Equation (4.7)] gives us

$$A_{0,1}(0) = A_{1,1}(0) = A_{1,0}(0) \qquad \text{and} \qquad A_{0,1}(1) = A_{1,1}(1) = A_{1,0}(1),$$

while [Izu18, Equation (4.8)] gives us $A_{1,1}(0) = A_{1,1}(1)$. Now, [Izu18, Equations (4.4) and (4.6)] tell us

$$A_{0,1}(0)A_{1,1}(1) + A_{1,1}(0)A_{1,0}(1) = 0.$$

Thus $A_{0,1}(g) = A_{1,1}(g) = A_{1,0}(g) = 0$ and hence $A_{0,0}(g) = -1/d$ by [Izu18, Equation (4.3)]. However, in this case we cannot satisfy [Izu18, Equation (4.9)]. Suppose instead that $\epsilon_1(1) = -1$. With this new 2-cocycle, [Izu18, Equation (4.7)] now gives us

$$A_{0,1}(0) = A_{1,1}(0) = A_{1,0}(0) \qquad \text{and} \qquad A_{0,1}(1) = -A_{1,1}(1) = A_{1,0}(1),$$

while [Izu18, Equation (4.8)] gives us $A_{1,1}(1) = -A_{1,1}(0)$. We then see by [Izu18, Equation (4.4)] that

$$A_{0,1}(0)A_{1,0}(0) + A_{1,1}(0)A_{1,1}(0) = 1 \implies A_{1,0}(0) = \pm\frac{1}{\sqrt{2}} = \pm\frac{1}{d-1},$$

and by [Izu18, Equation (4.9)] that

$$A_{0,0}(0)A_{1,0}(0)^2 = A_{1,0}(0)^2 + A_{1,0}(0)^3 \implies A_{0,0}(0) = 1 + A_{1,0}(0) = \frac{d-1\pm 1}{d-1}.$$

Finally, [Izu18, Equation (4.3)] allows us to deduce

$$A_{1,0}(0) = -\frac{1}{d-1},$$

whence

$$A(0) = \frac{1}{d-1}\begin{pmatrix} d-2 & -1 \\ -1 & -1 \end{pmatrix} \qquad \text{and} \qquad A(1) = \frac{1}{d-1}\begin{pmatrix} d-2 & -1 \\ -1 & 1 \end{pmatrix}.$$

This category is nothing but the even part of the type A_7 subfactor.

Remark 2.6. Suppose that $|G|$ is odd. Then [Izu18, Equation (4.1)] tells us that $\epsilon_h(g) = 1$, while [Izu18, Equation (4.2)] tells us that $\omega(g)$ does not depend on g. Moreover, $A_{h,k}(g)$ cannot depend on g by [Izu18, Equation (4.5)], and either $\omega = 1$ or $A_{0,0} = 0$ by [Izu18, Equation (4.7)]. In this case, [Izu18, Equations (4.1)–(4.9)] reduce to the following four equations.

$$A_{h,k} = A_{-k,h-k}\omega = A_{k-h,-h}\overline{\omega},$$

$$\sum_{h\in G} A_{h,0} = -\frac{\overline{\omega}}{d_\pm},$$

$$\sum_{h\in G} A_{h-g,k}A_{k,h-g'} = \delta_{g,g'} - \frac{\delta_{k,0}}{d_\pm},$$

$$\sum_{l\in G} A_{x+y,l}A_{-x,l+p}A_{-y,l+q} = A_{p+x,q+x+y}A_{q+y,p+x+y} - \frac{\delta_{x,0}\delta_{y,0}}{d_\pm}.$$

The first three equations above are precisely [EG17, Equations (4.7)–(4.9)]! In particular, to see that our third equation is equivalent to [EG17, Equation (4.9)], we simply make the change of variables $\hat{g} := g' - g$ and $\hat{h} := h - g'$, whence we obtain

$$\sum_{\hat{h}\in G} A_{\hat{h}+\hat{g},k}A_{k,\hat{h}} = \delta_{\hat{g},0} - \frac{\delta_{k,0}}{d_\pm}.$$

Similarly, using our first equation while making the change of variables $\hat{l} := l - x - y$, $\hat{p} := p + x + y$, $\hat{q} := q + x + y$, $\hat{x} := -x$ and $\hat{y} := -y$, our fourth equation becomes

$$\overline{\omega}\sum_{\hat{l}\in G} A_{\hat{l},\hat{x}+\hat{y}}A_{\hat{x},\hat{l}+\hat{p}}A_{\hat{y},\hat{l}+\hat{q}} = A_{\hat{y}+\hat{p},\hat{q}}A_{\hat{x}+\hat{q},\hat{p}} - \frac{\delta_{\hat{x},0}\delta_{\hat{y},0}}{d_\pm},$$

showing that it is equivalent to [EG17, Equation (4.11)].

3. The Leavitt Algebra Approach of Evans–Gannon

The important results are [EG17, Theorem 1 and Theorem 2].

Definition 3.1. (Leavitt Algebra). *Let $X \coloneqq (x_{ij})$ and $Y \coloneqq (y_{ij})$ be $m \times n$ and $n \times m$ matrices of symbols, respectively. The* Leavitt $\Bbbk$-algebra of type (m,n) *is the free associative unital $\Bbbk$-algebra*

$$\mathcal{L}_{\Bbbk}(m,n) \coloneqq \frac{\Bbbk[x_{ij}, y_{ij}]}{\langle XY = I_m, YX = I_n \rangle}.$$

In other words, it is the universal $\Bbbk$-algebra with generators

$$\{x_{ij} : 1 \le i \le m, 1 \le j \le n\} \sqcup \{y_{ij} : 1 \le i \le n, 1 \le j \le m\}$$

and Leavitt–Cuntz relations

$$\sum_{k=1}^{m} y_{ik}x_{kj} = \delta_{i,j} \qquad \text{and} \qquad \sum_{k=1}^{n} x_{ik}y_{kj} = \delta_{i,j},$$

for all suitable i, j.

Consider the Leavitt $\mathbb{C}$-algebra of type $(1,n)$, which we shall henceforth denote by $\mathcal{L}_n \coloneqq \mathcal{L}_{\mathbb{C}}(1,n)$. We have that $\mathcal{O}_n = C^*(\mathcal{L}_n)$, where $x_i = S_i$ and $y_i = S_i^*$. Let's think about what this means precisely. The Leavitt–Cuntz relations for $m = 1$ become

$$y_i x_j = \delta_{i,j} \qquad \text{and} \qquad \sum_{k=1}^{n} x_k y_k = 1.$$

We may endow $\mathcal{L}_n$ with the structure of a $*$-algebra by defining an involutive conjugate homogeneous antiautomorphism that sends $x_i \mapsto y_i$ and $y_i \mapsto x_i$. We further define

$$\|a\| \coloneqq \sup\{p(a) : p \text{ is a } C^*\text{-seminorm on } \mathcal{L}_n\}$$

for all $a \in \mathcal{L}_n$, where a C^*-seminorm is just a seminorm for which $p(a^*a) = p(a)^2$ and $p(ab) \le p(a)p(b)$. Note that $0 \le \|a\| \le 1$, as $1 = \|y_i x_i\| = \|x_i^2\|$ and hence $\|x_i\| = \|y_i\| = 1$ for all i. The condition $p(ab) \le p(a)p(b)$ ensures that $\mathcal{I} \coloneqq \{a \in \mathcal{L}_n : \|a\| = 0\}$ is an ideal in $\mathcal{L}_n$. Our C^*-seminorm then descends to a C^*-norm on the quotient $\mathcal{L}_n/\mathcal{I}$. The completion of $\mathcal{L}_n/\mathcal{I}$ with respect to this C^*-norm is known as the universal C^*-algebra of $\mathcal{L}_n$, denoted by $C^*(\mathcal{L}_n)$. This is precisely $\mathcal{O}_n$ by definition. We may therefore view $\mathcal{L}_n$ as the polynomial part of $\mathcal{O}_n$. Is this stuff really important?

Because $\mathcal{L}_n$ is a unital algebra over $\Bbbk$, its algebra endomorphisms define a strict preadditive $\Bbbk$-linear tensor category $\mathsf{End}(\mathcal{L}_n)$. The objects of $\mathsf{End}(\mathcal{L}_n)$ are algebra endomorphisms of $\mathcal{L}_n$, while the morphisms are intertwiners. In other words, a morphism $r \in \mathrm{Hom}_{\mathsf{End}(\mathcal{L}_n)}(\beta, \gamma)$ is an element $r \in \mathcal{L}_n$ for which $r\beta(x) = \gamma(x)r$ for all $x \in \mathcal{L}_n$, with composition being multiplication in $\mathcal{L}_n$. We endow $\mathsf{End}(\mathcal{L}_n)$ with a tensor product given on objects as $\beta \otimes \gamma \coloneqq \beta \circ \gamma$ for all $\beta, \gamma \in \mathrm{Ob}(\mathsf{End}(\mathcal{L}_n))$ and on morphisms as $r \otimes s \coloneqq r\beta(s) = \gamma(s)r \in \mathrm{Hom}_{\mathsf{End}(\mathcal{L}_n)}(\beta \circ \rho, \gamma \circ \sigma)$ for all $r \in \mathrm{Hom}_{\mathsf{End}(\mathcal{L}_n)}(\beta, \gamma)$ and $s \in \mathrm{Hom}_{\mathsf{End}(\mathcal{L}_n)}(\rho, \sigma)$. Note that this is indeed well-defined, since

$$r\beta(s)\beta(\rho(x)) = \gamma(s)r\beta(\rho(x)) = \gamma(s)\gamma(\rho(x))r = \gamma(s\rho(x))r = \gamma(\sigma(x)s)r = \gamma(\sigma(x))\gamma(s)r$$

for all $x \in \mathcal{L}_n$. We should be more general here.

Define rigid objects (see EGNO Definition 2.10.1).

Let $\mathcal{A}$ now be a collection of algebra endomorphisms of $\mathcal{L}_n$ that is closed under composition and that contains the identity. We denote by $\mathsf{End}(\mathcal{A})$ the subcategory of $\mathsf{End}(\mathcal{L}_n)$ restricted to $\mathcal{A}$. This subcategory is itself a $\mathbb{C}$-linear tensor category, and $\mathrm{End}_{\mathsf{End}(\mathcal{A})}(\mathrm{id}) = \mathbb{k}$. Suppose we now take its Karoubi envelope, $\overline{\mathsf{End}(\mathcal{A})}$. Its objects consist of pairs (p, β), where $\beta \in \mathcal{A}$ and $p \in \mathrm{End}_{\mathsf{End}(\mathcal{A})}(\beta)$ is an idempotent. The hom-spaces are of the form $\mathrm{Hom}_{\overline{\mathsf{End}(\mathcal{A})}}((p, \beta), (q, \gamma)) := q\mathrm{Hom}_{\mathsf{End}(\mathcal{A})}(\beta, \gamma)p$, with composition given once more by multiplication. We endow $\overline{\mathsf{End}(\mathcal{A})}$ with the structure of a tensor category by defining a monoidal product on objects by $(p, \beta) \otimes (q, \gamma) := (p \otimes q, \beta \otimes \gamma) = (p\beta(q), \beta \circ \gamma)$ and on morphisms by $(qrp) \otimes (q'r'p') := qrp\beta(q'r'p')$, for $qrp \in \mathrm{Hom}((p, \beta), (q, \gamma))$ and $q'r'p' \in \mathrm{Hom}((p', \beta'), (q', \gamma'))$. We also endow $\overline{\mathsf{End}(\mathcal{A})}$ with the structure of an additive category as follows. The objects are ordered n-tuples $((p_1, \beta_1), \ldots, (p_n, \beta_n)) =: (p_1, \beta_1) \oplus \cdots \oplus (p_n, \beta_n)$, while the hom-spaces are

$$\mathrm{Hom}(((p_1, \beta_1), \ldots, (p_n, \beta_n)), ((q_1, \gamma_1), \ldots, (q_m, \gamma_m)))$$
$$= \begin{pmatrix} q_1\mathrm{Hom}(\beta_1, \gamma_1)p_1 & \cdots & q_1\mathrm{Hom}(\beta_n, \gamma_1)p_n \\ \vdots & \ddots & \vdots \\ q_m\mathrm{Hom}(\beta_1, \gamma_m)p_1 & \cdots & q_m\mathrm{Hom}(\beta_n, \gamma_m)p_n \end{pmatrix}.$$

Composition of morphisms in this category corresponds to matrix multiplication. The monoidal product $((p_1, \beta_1), \ldots, (p_n, \beta_n)) \otimes ((q_1, \gamma_1), \ldots, (q_m, \gamma_m))$ is the direct sum of each $(p_i, \beta_i) \otimes (q_j, \gamma_j)$. On morphisms, the monoidal product acts as the Kronecker product, with $(ij, i'j')$th entry given by $q_i r_{ji} p_j \otimes q'_{i'} r'_{j'i'} p'_{j'}$. We will write $\overline{\mathsf{End}(\mathcal{A})}_{\oplus}$ for the Karoubi envelope $\overline{\mathsf{End}(\mathcal{A})}$ extended by direct sums.

Suppose as before that $\mathcal{A}$ is a collection of $\mathcal{L}_n$-endomorphisms that is closed under composition and contains the identity. Suppose that $\mathrm{Hom}(\beta, \gamma) = \mathrm{Hom}(\tilde{\beta}, \tilde{\gamma})$ in $\mathcal{L}_n$ for all $\beta, \gamma \in \mathcal{A}$, where $\tilde{\beta}(x) := \beta(x')'$, and moreover that these hom-spaces are all finite-dimensional. Then $\overline{\mathsf{End}(\mathcal{A})}_{\oplus}$ is a strict rigid semisimple $\mathbb{C}$-linear tensor category with finite-dimensional Hom-spaces by [EG17, Lemma 2].

Why do we do this? Well, for the Yang–Lee example it's not an issue, since we're working in $\mathcal{L}_2$. For example, given a sum $\rho \oplus (\rho^3 \oplus \rho^5)$, we can write it in the Leavitt algebra as

$$s\rho s' + t(s\rho^3 s' + t\rho^5 t')t'.$$

For an arbitrary category, however, we aren't so fortunate! In particular, we can only write sums whose number of simple summands k obeys $k \equiv 1 \pmod{m + n - 1}$ as genuine Leavitt algebra endomorphisms. This is exactly why we need to consider the idempotent completion.

Example 3.2. (Yang–Lee Category). Let $G = \{0\}$. Then [EG17, Equation (4.7)] demands that

$$A_{0,0} = \omega A_{0,0} = \overline{\omega} A_{0,0} \implies \omega = 1,$$

whence [EG17, Equation (4.8)] tells us that

$$A_{0,0} = -\frac{1}{d_\pm}.$$

The rest of [EG17, Equations (4.7)–(4.10)] are satisfied by these choices. Hence by [EG17, Theorem 2], we have two fusion categories for $G = \{0\}$; a unitary one with $\pm = +$ (the Fibonacci category) and a non-unitary one with $\pm = -$ (the Yang–Lee category).

Example 3.3. ($G = \mathbb{Z}/2\mathbb{Z}$). The equations we must satisfy for $|G|$ even are given in [Izu18] (is this true?). Adapting our argument from Example 2.5, we see that there is exactly one non-unitary Haagerup–Izumi category with $G = \mathbb{Z}/2\mathbb{Z}$. This corresponds to $\epsilon_h(g) = (-1)^{gh}$, $\omega(g) = 1$,

$$A(0) = \frac{1}{d-1}\begin{pmatrix} d & 1 \\ 1 & 1 \end{pmatrix} \qquad \text{and} \qquad A(1) = \frac{1}{d-1}\begin{pmatrix} d & 1 \\ 1 & -1 \end{pmatrix}.$$

4. Deconstruction

Let $\mathcal{C}$ be a near-group category with fusion rule (G, m) over an algebraically closed field $\Bbbk$ whose characteristic does not divide $n \coloneqq |G|$.

In this section, we assume that there exist algebra endomorphisms realizing the fusion rules of $\mathcal{C}$. We will make this precise momentarily. Our aim will be to show that these endomorphisms restrict to endomorphisms of the Leavitt algebra $\mathcal{L} \coloneqq \mathcal{L}_{n+m}$, and then to derive certain data / invariants.

Let us first write our fusion rules in the language of algebra endomorphisms. Suppose there exist endomorphisms $\{\alpha_g\}_{g\in G}$ and ρ of some algebra A such that

$$[\alpha_g \circ \alpha_h] = [\alpha_{gh}], \tag{4.1}$$

$$[\alpha_g \circ \rho] = [\rho] = [\rho \circ \alpha_g], \tag{4.2}$$

where isomorphisms are given by invertible intertwiners. Let $\alpha'_g \in [\alpha_g]$ such that $\alpha'_g \circ \rho = \rho'$. Then there exists some invertible $\omega_g \in \mathrm{Hom}_{\mathsf{End}(A)}(\rho', \rho)$ for which $\alpha'_g(\rho(x)) = \omega_g^{-1}\rho(x)\omega_g$. We may therefore choose representatives $\alpha_g : x \mapsto \omega_g \alpha'_g(x)\omega_g^{-1}$ such that $\alpha_g \circ \rho = \rho$. Moreover, there must exist some $\mu_{g,h} \in \mathrm{Hom}_{\mathsf{End}(A)}(\alpha_{gh}, \alpha_g \circ \alpha_h)$ such that $\alpha_g(\alpha_h(x)) = \mu_{g,h}\alpha_{gh}(x)\mu_{g,h}^{-1}$. Thus

$$\rho(x) = \alpha_g(\alpha_h(\rho(x))) = \mu_{g,h}\alpha_{gh}(\rho(x))\mu_{g,h}^{-1} = \mu_{g,h}\rho(x)\mu_{g,h}^{-1}. \tag{4.3}$$

It follows that $\mu_{g,h} \in \mathrm{Hom}_{\mathsf{End}(A)}(\rho, \rho)$. Since we want ρ to be simple, we assume $\mathrm{Hom}_{\mathsf{End}(A)}(\rho, \rho)$ to be 1-dimensional. Thus $\mu_{g,h}$ is a scalar, whence $\alpha_g \circ \alpha_h = \alpha_{gh}$ for all $g, h \in G$. With such choices, we may only expect $\rho(\alpha_g(x)) = U(g)\rho(x)U(g)^{-1}$, for some invertible $U(g) \in \mathrm{Hom}_{\mathsf{End}(A)}(\rho, \rho \circ \alpha_g)$.

We therefore assume that there exist endomorphisms $\{\alpha_g\}_{g\in G}$ and ρ of some algebra A, together with $s_g, s'_g, t_i, t'_i \in A$ and invertibles $U(g) \in A^\times$ such that

$$\alpha_g(\alpha_h(x)) = \alpha_{gh}(x), \tag{4.4}$$

$$\alpha_g(\rho(x)) = \rho(x), \qquad \rho(\alpha_g(x)) = U(g)\rho(x)U(g)^{-1}, \tag{4.5}$$

$$\rho(\rho(x)) = \sum_{g\in G} s_g\alpha_g(x)s'_g + \sum_{i=1}^{m} t_i\rho(x)t'_i \tag{4.6}$$

for all $x \in A$, where

$$\sum_{g\in G} s_g s'_g + \sum_{i=1}^{m} t_i t'_i = 1, \qquad s'_g s_h = \delta_{g,h}, \qquad t'_i t_j = \delta_{i,j} \qquad \text{and} \qquad s'_g t_i = 0 = t'_i s_g, \tag{4.7}$$

for all $g, h \in G$ and $i, j \in \{1, \dots, m\}$. For the sake of convenience, we will define

$$S \coloneqq \sum_{g\in G} s_g s'_g \qquad \text{and} \qquad T \coloneqq \sum_{i=1}^{m} t_i t'_i, \tag{4.8}$$

where $S+T=1$ by definition. For $\{\alpha_g\}_{g\in G}$ and ρ to be (absolutely) simple, we assume also that

$$\mathrm{Hom}_{\mathsf{End}(A)}(\alpha_g,\alpha_h)=\Bbbk\delta_{g,h}, \tag{4.9}$$

$$\mathrm{Hom}_{\mathsf{End}(A)}(\alpha_g,\rho),=\{0\}=\mathrm{Hom}_{\mathsf{End}(A)}(\rho,\alpha_g) \tag{4.10}$$

$$\mathrm{Hom}_{\mathsf{End}(A)}(\rho,\rho)=\Bbbk, \tag{4.11}$$

for all $g,h\in G$. Then we have the following result.

Lemma 4.1. *Let $\{\alpha_g\}_{g\in G}$ and ρ be endomorphisms of A satisfying our prior assumptions. Then*

$$\mathrm{Hom}_{\mathsf{End}(A)}(\alpha_g,\rho^2)=\mathrm{span}_\Bbbk\{s_g\}, \tag{4.12}$$

$$\mathrm{Hom}_{\mathsf{End}(A)}(\rho^2,\alpha_g)=\mathrm{span}_\Bbbk\{s'_g\}, \tag{4.13}$$

$$\mathrm{Hom}_{\mathsf{End}(A)}(\rho,\rho^2)\ \ =\mathrm{span}_\Bbbk\{t_1,\dots,t_m\}, \tag{4.14}$$

$$\mathrm{Hom}_{\mathsf{End}(A)}(\rho^2,\rho)\ \ =\mathrm{span}_\Bbbk\{t'_1,\dots,t'_m\}, \tag{4.15}$$

$$\mathrm{Hom}_{\mathsf{End}(A)}(\rho^2,\rho^2)\ =\mathrm{span}_\Bbbk\{s_gs'_g\}_{g\in G}\sqcup\{t_it'_j\}_{i,j=1}^m, \tag{4.16}$$

for all $g\in G$.

Proof. From the fusion rules, $\rho^2(x)s_g=s_gx$, $s'_g\rho^2(x)=xs'_g$, $\rho^2(x)t_i=t_i\rho(x)$ and $t'_i\rho^2(x)=\rho(x)t'_i$, for all $x\in A$, $g\in G$ and $i\in\{1,\dots,m\}$. In other words, $s_g\in\mathrm{Hom}_{\mathsf{End}(A)}(\alpha_g,\rho^2)$, $s'_g\in\mathrm{Hom}_{\mathsf{End}(A)}(\rho^2,\alpha_g)$, $t_i\in\mathrm{Hom}_{\mathsf{End}(A)}(\rho,\rho^2)$ and $t'_i\in\mathrm{Hom}_{\mathsf{End}(A)}(\rho^2,\rho)$. Composing these tells us also that we must have $s_gs'_g, t_it'_j\in\mathrm{Hom}_{\mathsf{End}(A)}(\rho^2,\rho^2)$. Suppose $r\in\mathrm{Hom}_{\mathsf{End}(A)}(\alpha_g,\rho^2)$. Then $s'_hr\in\mathrm{Hom}_{\mathsf{End}(A)}(\alpha_g,\alpha_h)=\Bbbk\delta_{g,h}$ and $t'_ir\in\mathrm{Hom}_{\mathsf{End}(A)}(\alpha_g,\rho)=\{0\}$, whence

$$r=(S+T)r=s_gs'_gr\in\Bbbk s_g. \tag{4.17}$$

Similar arguments give us Equations (4.13), (4.14), (4.15) and (4.16). This completes the proof. ■

We will find it convenient to denote $\mathcal{K}:=\mathrm{span}_\Bbbk\{t_1,\dots,t_m\}$ and $\mathcal{K}':=\mathrm{span}_\Bbbk\{t'_1,\dots,t'_m\}$, and let

$$\mathcal{K}^p\mathcal{K}'^q:=\mathrm{span}_\Bbbk\{t_{i_1}t_{i_2}\cdots t_{i_p}t'_{j_1}t'_{j_2}\cdots t'_{j_q}\}_{i_1,\dots,i_p,j_1,\dots,j_q=1}^m, \tag{4.18}$$

for $p,q\in\mathbb{N}$. In other words, $\mathcal{K}^p\mathcal{K}'^q\cong\mathcal{K}^{\otimes p}\otimes(\mathcal{K}')^{\otimes q}$. We also note that $\mathcal{K}\mathcal{K}'=\mathrm{End}(\mathcal{K})$. Certainly $\mathcal{K}\mathcal{K}'\subseteq\mathrm{End}(\mathcal{K})$, since $t_it'_j\mathcal{K}\subseteq\mathcal{K}$. The converse direction comes by comparing dimensions. A dual argument with $\mathcal{K}'t_it'_j\subseteq\mathcal{K}'$ shows also that $\mathcal{K}\mathcal{K}'=\mathrm{End}(\mathcal{K}')$.

Lemma 4.2. *The automorphism α_g admits the dual $\alpha_{g^{-1}}$ and restricts to an automorphism of $\mathcal{L}$, for all $g\in G$. In particular, there exists a representation $V:G\to\mathrm{End}(\mathcal{K})$ such that*

$$\alpha_g(x)=V(g)x\qquad\text{and}\qquad\alpha_g(y)=yV(g)^{-1}, \tag{4.19}$$

for all $g\in G$, $x\in\mathcal{K}$ and $y\in\mathcal{K}'$.

Proof. We know immediately that $\alpha_{g^{-1}}$ is a left and a right dual for α_g, these are linear maps and

$$\mathrm{Hom}_{\mathsf{End}(A)}(\mathrm{id},\alpha_g\circ\alpha_{g^{-1}})=\mathrm{Hom}_{\mathsf{End}(A)}(\alpha_e,\alpha_e)=\Bbbk=\mathrm{Hom}_{\mathsf{End}(A)}(\alpha_{g^{-1}}\circ\alpha_g,\mathrm{id}), \tag{4.20}$$

From the fact that α_g is an endomorphism, we also know that

$$\alpha_g(s_h) \in \mathrm{Hom}_{\mathsf{End}(A)}(\alpha_{gh}, \rho^2) = \Bbbk s_{gh}, \tag{4.21}$$

$$\alpha_g(s'_h) \in \mathrm{Hom}_{\mathsf{End}(A)}(\rho^2, \alpha_{gh}) = \Bbbk s'_{gh}, \tag{4.22}$$

$$\alpha_g(x) \in \mathcal{K}, \tag{4.23}$$

$$\alpha_g(y) \in \mathcal{K}', \tag{4.24}$$

for all $g, h \in G$, $x \in \mathcal{K}$ and $y \in \mathcal{K}'$. In particular, there must exist an element $V(g) \in \mathcal{K}\mathcal{K}'$ satisfying $\alpha_g(x) = V(g)x$ for all $x \in \mathcal{K}$, defining a representation $V : G \to \mathrm{End}(\mathcal{K})$. In fact, $1 = \alpha_g(t'_i t_i) = \alpha_g(t'_i)V(g)t_i$ tells us that $\alpha_g(y) = yV(g)^{-1}$ for all $y \in \mathcal{K}'$, as both $V(g)$ and t_i have unique left inverses. This completes the proof. ∎

Observe that

$$U(g)U(h)\rho(x)U(h)^{-1}U(g)^{-1} = \rho(\alpha_g(\alpha_h(x))) = \rho(\alpha_{gh}(x)) = U(gh)\rho(x)U(gh)^{-1} \tag{4.25}$$

$$\implies U(gh)^{-1}U(g)U(h)\rho(x) = \rho(x)U(gh)^{-1}U(g)U(h). \tag{4.26}$$

Thus $U(gh)^{-1}U(g)U(h)$ intertwines ρ and hence lies in $\Bbbk^\times$. In particular, $U : G \mapsto \mathrm{Hom}_{\mathsf{End}(A)}(\rho^2, \rho^2)$ defines a projective representation of G. Now, $\alpha_g(\alpha_e(x)) = \alpha_g(x)$ for all $x \in A$, so $\alpha_e = \mathrm{id}$ and $U(g)s_e \in \mathrm{Hom}_{\mathsf{End}(A)}(\mathrm{id}, \rho^2)$. We shall choose to rescale $U(g)$ such that $U(g)s_e = s_e$ for all $g \in G$, making U a genuine representation of G. Moreover, we have from Equation (4.16) that

$$U(g) = \sum_{h \in G} \chi_h(g) s_h s'_h + U_{\mathcal{K}}(g) \tag{4.27}$$

for $\chi_h : G \to \Bbbk$ and $U_{\mathcal{K}}(g) \in \mathrm{span}\{t_i t'_j\}_{i,j=1}^m = \mathcal{K}\mathcal{K}'$. In particular, $U_{\mathcal{K}} : G \to \mathrm{End}(\mathcal{K})$ defines a representation as the projection of $U(g)$ onto $\mathcal{K}\mathcal{K}'$, and $\chi_h \in \mathrm{Hom}(G, \Bbbk^\times)$. Indeed, if $\chi_h(g) = 0$, then $U(g)s_h = 0 = s'_h U(g)$; this would imply $s_h = 0 = s'_h$ by the invertibility of $U(g)$, which is nonsense. Moreover, the fact that U and $U_{\mathcal{K}}$ are group homomorphisms shows $\chi_h(g_1 g_2) = \chi_h(g_1)\chi_h(g_2)$. In other words, χ_h is a character of G.

Since $s_h \in \mathrm{Hom}_{\mathsf{End}(A)}(\alpha_h, \rho^2)$ and since α_g is an automorphism, $\alpha_g(s_h) \in \Bbbk^\times s_{gh}$. Similarly, because $s'_h \in \mathrm{Hom}_{\mathsf{End}(A)}(\rho^2, \alpha_h)$, we have that $\alpha_g(s'_h) \in \Bbbk^\times s'_{gh}$. In what follows, we will choose s_g and s'_g such that $\alpha_g(s_e) = s_g$ and $\alpha_g(s'_e) = s'_g$, for all $g \neq e$. Then from $\alpha_g \circ \alpha_h = \alpha_{gh}$, it follows that $\alpha_g(s_h) = s_{gh}$.

The above choice can definitely be removed. But we'll have to recheck our proofs.

Consider the linear maps $j_1 : \mathcal{K}' \to \mathcal{K}$ and $j'_1 : \mathcal{K} \to \mathcal{K}'$ defined by

$$j_1 : y \mapsto y\rho(s_e) \qquad \text{and} \qquad j'_1 : x \mapsto \rho(s'_e)x, \tag{4.28}$$

together with the linear maps $j_2 : \mathcal{K}' \to \mathcal{K}$ and $j'_2 : \mathcal{K} \to \mathcal{K}'$ defined by

$$j_2 : y \mapsto \rho(y)s_e \qquad \text{and} \qquad j'_2 : x \mapsto s'_e\rho(x). \tag{4.29}$$

These allow us to state the following.

Lemma 4.3. *The endomorphism ρ restricts to an endomorphism of $\mathcal{L}$. In particular, there exists some $a \in \Bbbk$ such that*

$$\rho(s_e) = a\sum_{h\in G} s_h + \sum_{i=1}^{m} t_i j_1(t_i'), \tag{4.30}$$

$$\rho(s_e') = a\sum_{h\in G} s_h' + \sum_{i=1}^{m} j_1'(t_i)t_i', \tag{4.31}$$

as well as linear maps and $l_h : \mathcal{K} \to \mathcal{K}$, $l : \mathcal{K} \to \mathcal{K}^2\mathcal{K}'$, $l_h' : \mathcal{K}' \to \mathcal{K}'$ and $l' : \mathcal{K}' \to \mathcal{K}(\mathcal{K}')^2$ such that

$$\rho(x) = \sum_{h\in G} s_h \alpha_h(j_2'(x)) + \sum_{h\in G} \alpha_h(l_e(x)) s_h s_h' + l(x), \tag{4.32}$$

$$\rho(y) = \sum_{h\in G} \alpha_h(j_2(y)) s_h' + \sum_{h\in G} s_h s_h' \alpha_h(l_e'(y)) + l'(y), \tag{4.33}$$

for all $x \in \mathcal{K}$ and $y \in \mathcal{K}'$.

Proof. Observe that $s_e'\rho(s_e) \in \mathrm{Hom}_{\mathsf{End}(A)}(\rho,\rho) = \Bbbk$ and $\rho(s_e')s_e \in \mathrm{Hom}_{\mathsf{End}(A)}(\rho,\rho) = \Bbbk$. Let

$$a := s_e'\rho(s_e) \qquad \text{and} \qquad a' := \rho(s_e')s_e. \tag{4.34}$$

We first note that, by Equation (4.6), we have

$$a = \rho(a) = \rho(s_e')\rho^2(s_e) = \sum_{g\in G} \rho(s_e') s_g s_g s_g' + \sum_{i=1}^{m} \rho(s_e') t_i \rho(s_e) t_i' = a'S + aT. \tag{4.35}$$

It follows that $a' = a$ and hence

$$\rho(s_e) = (S+T)\rho(s_e) = a\sum_{h\in G} s_h + \sum_{i=1}^{m} t_i j_1(t_i') \in \mathcal{L}, \tag{4.36}$$

$$\rho(s_e') = \rho(s_e')(S+T) = a\sum_{h\in G} s_h' + \sum_{i=1}^{m} j_1'(t_i)t_i' \in \mathcal{L}. \tag{4.37}$$

Similarly, $t_i'\rho(t_j) \in \mathrm{Hom}_{\mathsf{End}(A)}(\rho^2,\rho^2)$ and $\rho(t_j')t_i \in \mathrm{Hom}_{\mathsf{End}(A)}(\rho^2,\rho^2)$. Thus by Equation (4.16), we have $T\rho(x) \in \mathcal{K}\mathrm{Hom}_{\mathsf{End}(A)}(\rho^2,\rho^2)$ for all $x \in \mathcal{K}$, whence there must exist linear maps $l_h : \mathcal{K} \to \mathcal{K}$ and $l : \mathcal{K} \to \mathcal{K}^2\mathcal{K}'$ for which

$$T\rho(x) = \sum_{h\in G} l_h(x) s_h s_h' + l(x). \tag{4.38}$$

Dually, since $\rho(y)T \in \mathrm{Hom}_{\mathsf{End}(A)}(\rho^2,\rho^2)\mathcal{K}'$ for all $y \in \mathcal{K}'$, we must also have linear maps $l_h' : \mathcal{K}' \to \mathcal{K}'$ and $l' : \mathcal{K}' \to \mathcal{K}(\mathcal{K}')^2$ such that

$$\rho(y)T = \sum_{h\in G} s_h s_h' l_h'(y) + l'(y). \tag{4.39}$$

Because $\alpha_g \circ \rho = \rho$, we must have

$$T\alpha_g(\rho(x)) = \sum_{h\in G} \alpha_g(l_h(x)) s_{gh} s_{gh}' + \alpha_g(l(x)) = T\rho(x); \tag{4.40}$$

thus $l_g = \alpha_g \circ l_e$, $\alpha_g \circ l = l$, $l'_g = \alpha_g \circ l'_e$, $\alpha_g \circ l' = l'$. The result follows. This completes the proof. ■

Remark 4.4. While we were not able to show that ρ is rigid a priori (that is, that $a \neq 0$), Lemma 4.1 can show that it is at least *weakly* self-dual ([BK01, Definition 5.3.4]). In the unitary setting, this does happen to be enough for rigidity; see [EP26, Remark 3.2], for instance. Even if we could show that it is self-dual, however, we will find it important to work with a pivotal structure. In principle we could define a partial pivotal structure on only the simple objects and ρ^2, but this too seems infeasible.

Rather than continuing to work in $\mathsf{End}(A)$, let's work instead in the monoidal subcategory $\mathcal{D}$ given by restricting $\mathsf{End}(\mathcal{L})$ to the closure of $\{\rho\} \sqcup \{\alpha_g\}_{g\in G}$ under composition. Then $\mathcal{D}$ is a full monoidal subcategory of $\mathcal{C}$ by Lemma 4.1. In particular, it inherits both rigidity and a pivotal structure ψ from $\mathcal{C}$. Because it is rigid, the left quantum trace Tr^L is genuinely a trace by [EGNO16, Proposition 4.7.3], and we can use the pivotal structure ψ to define the quantum dimension $\mathrm{Dim}(-) := \mathrm{Tr}^L(\psi_-)$. In particular, by Remark 8.3, we know that $\psi_\rho =: \nu \in \{-1, 1\}$. Note that $\mathcal{D}$ is only preadditive in general; the only direct sum that we know exists is

$$\rho^2 = \bigoplus_{g\in G} \alpha_g \oplus \bigoplus_{i=1}^{m} \rho, \tag{4.41}$$

which follows directly from the fusion rules. Need a reference to Thornton.

In $\mathcal{D}$, we have that $\rho^* = \rho = {}^*\rho$. From Lemma 4.1, we see that $\mathrm{coev}_\rho = bs_e$ and $\mathrm{ev}_\rho = b's'_e$, for $b, b' \in \Bbbk^\times$. It then follows from [EGNO16, Proposition 4.7.3] that

$$\mathrm{Tr}^L_{\rho^2}(S+T) = \mathrm{Tr}^L_\rho(1)\mathrm{Tr}^L_\rho(1) = \mathrm{ev}_\rho \mathrm{coev}_\rho \mathrm{ev}_\rho \mathrm{coev}_\rho = (b'b)^2, \tag{4.42}$$

From the fact that $\rho^2 = \bigoplus_{g\in G} \alpha_g \oplus \bigoplus_{i=1}^{m} \rho$, we then have

$$\mathrm{Tr}^L_{\rho^2}(S+T) = \sum_{g\in G} \mathrm{Tr}^L_{\alpha_g}(1) + \sum_{i=1}^{m} \mathrm{Tr}^L_\rho(1) = n + mb'b. \tag{4.43}$$

Thus $(b'b)^2 = n + mb'b$ and hence $b'b = d_\pm$, where

$$d_\pm := \frac{m \pm \sqrt{m^2+4n}}{2}. \tag{4.44}$$

Finally, the zig-zag identity gives us

$$d_\pm a = b's'_e\rho(bs_e) = \mathrm{ev}_\rho \rho(\mathrm{coev}_\rho) = 1, \tag{4.45}$$

and so $a = 1/d_\pm$.

Corollary 4.5. *We have*

$$\rho(s'_e)t_i t'_j \rho(s_e) = \frac{\delta_{i,j}}{d_\pm}, \tag{4.46}$$

for all $i, j \in \{1, \ldots, m\}$.

Proof. By Equation (4.30) and Equation (4.31), together with the definition of $d_\pm$, we have

$$1 = \rho(s_e')\rho(s_e) = \frac{n}{d_\pm^2} + \sum_{i=1}^m \rho(s_e')t_it_i'\rho(s_e) \implies \sum_{i=1}^m \rho(s_e')t_it_i'\rho(s_e) = \frac{m}{d_\pm}. \tag{4.47}$$

Now, for $x \in \mathrm{Hom}_{\mathsf{End}(A)}(\rho^2, \rho^2)$,

$$\mathrm{Tr}^L_{\rho^2}(x) := \mathrm{ev}_\rho \rho(\mathrm{ev}_\rho) x \rho(\mathrm{coev}_\rho)\mathrm{coev}_\rho = d_\pm^2 s_e' \rho(s_e') x \rho(s_e) s_e = d_\pm^2 \rho(s_e') x \rho(s_e). \tag{4.48}$$

The final equality follows from the fact that $\rho(s_e')x\rho(s_e) \in \Bbbk$, as it is in the 1-dimensional space $\mathrm{Hom}_{\mathcal{D}}(\rho, \rho)$, together with $s_e's_e = 1$. Because $M_m(\Bbbk) \cong \mathcal{K}\mathcal{K}'$ admits a unique tracial state given by the matrix trace Tr, we know it must be related to the trace $\mathrm{Tr}^L_{\rho^2}$ by a scalar. Since $\mathrm{Tr}(T) = m$ while $\mathrm{Tr}^L_{\rho^2}(T) = md$ by our previous observation, we have for all $x \in \mathcal{K}\mathcal{K}' \subset \mathrm{Hom}_{\mathsf{End}(A)}(\rho^2, \rho^2)$ that

$$\rho(s_e')x\rho(s_e) = \frac{1}{d_\pm^2}\mathrm{Tr}^L_{\rho^2}(x) = \frac{1}{d_\pm}\mathrm{Tr}(x) = \frac{1}{d_\pm}\sum_{i=k}^m t_k' x t_k, \tag{4.49}$$

whence the result follows by considering $x = t_it_j'$. This completes the proof. ■

Suppose we define maps

$$\varphi := \frac{d_\pm}{\nu} j_2 \circ j_1' \in \mathcal{K}\mathcal{K}' \qquad \text{and} \qquad \varphi' := \frac{d_\pm}{\nu} j_2' \circ j_1 \in \mathcal{K}\mathcal{K}' \tag{4.50}$$

Then we have the following.

Lemma 4.6. *The maps φ and φ' are invertible with $\varphi^{-1} = \frac{d_\pm}{\nu} j_1 \circ j_2'$ and $\varphi'^{-1} = \frac{d_\pm}{\nu} j_1' \circ j_2$, respectively. Moreover, we have*

$$\varphi^3(x) = x, \tag{4.51}$$

$$\varphi(\alpha_g(\varphi^{-1}(x))) = U_{\mathcal{K}}(g)x, \tag{4.52}$$

$$\varphi^{-1}(\alpha_g(\varphi(x))) = \rho(U(g))xU(g)^{-1}, \tag{4.53}$$

for all $x \in \mathcal{K}$ and $g \in G$, and

$$\varphi'^3(y) = y, \tag{4.54}$$

$$\varphi'(\alpha_g(\varphi'^{-1}(y))) = yU_{\mathcal{K}}(g)^{-1}, \tag{4.55}$$

$$\varphi'^{-1}(\alpha_g(\varphi'(y))) = U(g)y\rho(U(g)^{-1}), \tag{4.56}$$

for all $y \in \mathcal{K}'$ and $g \in G$.

Proof. We begin by showing that $\varphi^{-1} = \frac{d_\pm}{\nu} j_1 \circ j_2'$ is indeed an inverse for φ. Using the intertwiner relation together with the zig-zag identity $\rho(s_e')s_e = 1/d_\pm$, we obtain

$$\begin{aligned}\varphi(\varphi^{-1}(x)) &= d_\pm^2 \rho^2(s_e')\rho(s_e')\rho^2(x)\rho^2(s_e)s_e = d_\pm^2 \rho^2(s_e')\rho(s_e')\rho^2(x)s_es_e \\ &= d_\pm^2 \rho^2(s_e')\rho(s_e')s_exs_e = d_\pm \rho^2(s_e')xs_e = d_\pm x\rho(s_e')s_e = x.\end{aligned}$$

A similar process using $s'_e\rho(s_e) = 1/d_\pm$ yields

$$\begin{aligned}\varphi^{-1}(\varphi(x)) &= d_\pm^2 s'_e\rho^3(s'_e)\rho^2(x)\rho(s_e)\rho(s_e) = d_\pm^2\rho(s'_e)s'_e\rho^2(x)\rho(s_e)\rho(s_e)\\ &= d_\pm^2\rho(s'_e)xs'_e\rho(s_e)\rho(s_e) = d_\pm\rho(s'_e)x\rho(s_e) = d_\pm\rho(s'_e)\rho^2(s_e)x = x.\end{aligned}$$

In order to show that $\varphi^3(x) = x$, recall that the right dual to $x \in \mathcal{K}$ is given by

$$^*x = d_\pm^{\frac{3}{2}}\rho(s'_e)\rho^2(s'_e)\rho(x)s_e \in \mathcal{K}',$$

while the right dual to $y \in \mathcal{K}'$ is given by

$$^*y = d_\pm^{\frac{3}{2}}\rho^2(s'_e)\rho^2(y)\rho(s_e)s_e \in \mathcal{K}.$$

Thus, for $x \in \mathcal{K}$, we have

$$^{**}x = d_\pm^3\rho^2(s'_e)\rho^3(s'_e)\rho^4(s'_e)\rho^3(x)\rho^2(s_e)\rho(s_e)s_e = \nu\varphi^3(x).$$

By naturality of the pivotal structure, we then have that $^{**}x = \psi_\rho\rho(\psi_\rho)x\psi_\rho^{-1} = \nu x$. Thus $\varphi^3(x) = x$. Now, using the normalization relation $U(g)^{-1}s_e = s_e$ and the definition of morphisms,

$$\begin{aligned}\varphi(\alpha_g(\varphi^{-1}(x))) &= d_\pm^2\rho^2(s'_e)\rho(\alpha_g(s'_e\rho(x)\rho(s_e)))s_e = d_\pm^2\rho^2(s'_e)U(g)\rho(s'_e)\rho^2(x)\rho^2(s_e)U(g)^{-1}s_e\\ &= d_\pm^2U(g)\rho^2(s'_e)\rho(s'_e)\rho^2(x)\rho^2(s_e)s_e = d_\pm^2U(g)\rho^2(s'_e)\rho(s'_e)s_exs_e\\ &= d_\pm U(g)\rho^2(s'_e)xs_e = d_\pm U(g)x\rho(s'_e)s_e = U(g)x = U_{\mathcal{K}}(g)x.\end{aligned}$$

Finally, using the normalization relation $s'_e = s'_eU(g)$, we have

$$\begin{aligned}\varphi^{-1}(\alpha_g(\varphi(x))) &= d_\pm^2s'_e\rho(\alpha_g(\rho^2(s'_e)\rho(x)s_e))\rho(s_e) = d_\pm^2s'_eU_g\rho^3(s'_e)\rho^2(x)\rho(s_e)U_g^{-1}\rho(s_e)\\ &= d_\pm^2s'_e\rho^3(s'_e)\rho^2(x)\rho(s_e)U_g^{-1}\rho(s_e) = d_\pm^2\rho(s'_e)s'_e\rho^2(x)\rho(s_e)U_g^{-1}\rho(s_e)\\ &= d_\pm^2\rho(s'_e)xs'_e\rho(s_e)U_g^{-1}\rho(s_e) = d_\pm\rho(s'_eU_{g^{-1}}^{-1})xU_{g^{-1}}\rho(s_e)\\ &= d_\pm\rho(s'_e)\rho(U_{g^{-1}}^{-1})x\rho(\alpha_{g^{-1}}(s_e))U_{g^{-1}} = d_\pm\rho(s'_e)\rho(U_{g^{-1}}^{-1})\rho^2(\alpha_{g^{-1}}(s_e))xU_{g^{-1}}\\ &= d_\pm\rho(s'_e)\rho(U_{g^{-1}}^{-1})\rho(U_{g^{-1}})\rho^2(s_e)\rho(U_{g^{-1}}^{-1})xU_{g^{-1}} = \rho(U_{g^{-1}}^{-1})xU_{g^{-1}}.\end{aligned}$$

The dual results for φ' follow similarly, but with $y^{**} = \nu\varphi'^3(y)$. This completes the proof. ■

The result $\varphi V(g)\varphi^{-1}x = U_{\mathcal{K}}(g)x$ can be rewritten as $\varphi V(g) = U_{\mathcal{K}}(g)\varphi$. In other words, the representations $U_{\mathcal{K}}$ and V are equivalent and intertwined by φ. Dually, $y\varphi'^{-1}V(g)^{-1}\varphi' = yU_{\mathcal{K}}(g)^{-1}$ and hence φ' intertwines $U_{\mathcal{K}}$ and V as representations on $\mathrm{End}(\mathcal{K}')$.

An immediate consequence of Lemma 4.6 is that $l_e = \nu\varphi$ and $l'_e = \nu\varphi'$. Indeed,

$$\begin{aligned}T\rho(\nu\varphi^{-1}(x))S &= T\rho(\nu\varphi^{-1}(x))\sum_{h\in G}s_hs'_h = Td_\pm\rho(s'_e)\rho^2(x)\rho^2(s_e)\sum_{h\in G}s_hs'_h\\ &= Td_\pm\rho(s'_e)\rho^2(x)\sum_{h\in G}s_hs_hs'_h = Td_\pm\rho(s'_e)\sum_{h\in G}s_h\alpha_h(x)s_hs'_h\\ &= T\sum_{h\in G}\alpha_h(x)s_hs'_h = \sum_{h\in G}\alpha_h(x)s_hs'_h.\end{aligned}$$

Thus $\alpha_h(l_e(\varphi^{-1}(x))) = \alpha_h(x)$, whence composing both sides with $\alpha_{h^{-1}}$ gives us the desired result. The dual result follows similarly.

These results allow us to rewrite the equations for ρ in Lemma 4.3 as

$$\rho(s_e) = \frac{1}{d_\pm}\sum_{h\in G} s_h + \sum_{i=1}^{m} t_i j_1(t_i'), \tag{4.57}$$

$$\rho(s_e') = \frac{1}{d_\pm}\sum_{h\in G} s_h' + \sum_{i=1}^{m} j_1'(t_i)t_i', \tag{4.58}$$

and

$$\rho(x) = \sum_{h\in G} s_h\alpha_h(j_2'(x)) + d_\pm\sum_{h\in G}\alpha_h(j_2(j_1'(x)))s_hs_h' + l(x), \tag{4.59}$$

$$\rho(y) = \sum_{h\in G} \alpha_h(j_2(y))s_h' + d_\pm\sum_{h\in G} s_hs_h'\alpha_h(j_2'(j_1(y))) + l'(y), \tag{4.60}$$

for all $x \in \mathcal{K}$ and $y \in \mathcal{K}'$.

Lemma 4.7. *We have*

$$\chi_h(g)V(h)U_{\mathcal{K}}(g) = U_{\mathcal{K}}(g)V(h),$$
$$l(\alpha_g(x)) = U(g)l(x)U(g)^{-1},$$
$$l'(\alpha_g(y)) = U(g)l'(y)U(g)^{-1},$$

for all $x \in \mathcal{K}$, $y \in \mathcal{K}'$ and $g, h \in G$.

Proof. These relations follow from $\rho(\alpha_g(x)) = U(g)\rho(x)U(g)^{-1}$. By Lemma 4.6 and Equation (4.59),

$$\begin{aligned}\rho(\alpha_g(x)) &= \sum_{h\in G} s_hs_h'\rho(\alpha_g(x)) + \nu\sum_{h\in G} V(h)\varphi(\alpha_g(x))s_hs_h' + l(\alpha_g(x))\\ &= \sum_{h\in G} s_hs_h'U(g)\rho(x)U(g)^{-1} + \nu\sum_{h\in G} V(h)U_{\mathcal{K}}(g)\varphi(x)s_hs_h' + l(\alpha_g(x)).\end{aligned}$$

Meanwhile,

$$\begin{aligned}U(g)\rho(x)U(g)^{-1} &= \sum_{h\in G} U(g)s_hs_h'\rho(x)U(g)^{-1} + \nu\sum_{h\in G} U(g)V(h)\varphi(x)s_hs_h'U(g)^{-1} + U(g)l(x)U(g)^{-1}\\ &= \sum_{h\in G} s_hs_h'U(g)\rho(x)U(g)^{-1} + \nu\sum_{h\in G}\chi_h(g)^{-1}U_{\mathcal{K}}(g)V(h)\varphi(x)s_hs_h' + U(g)l(x)U(g)^{-1}.\end{aligned}$$

The final equation follows from the dual results. This completes the proof. ■

Corollary 4.8. *The map χ is a symmetric bicharacter. In particular,*

$$\chi_h(g) = \chi_g(h),$$
$$\alpha_h(U(g)) = \chi_h(g)^{-1}U(g)$$

for all $g, h \in G$.

Proof. Observe that

$$\rho(\alpha_g(x)) = \alpha_h(\rho(\alpha_g(x))) = \alpha_h(U(g)\rho(x)U(g)^{-1}) = \alpha_h(U(g))\rho(x)\alpha_h(U(g)^{-1}).$$

It follows that $\alpha_h(U(g))$ is a scalar multiple of $U(g)$. In particular,

$$\alpha_h(U(g))s_e = \alpha_h(U(g)s_{h^{-1}}) = \chi_{h^{-1}}(g)\alpha_h(s_{h^{-1}}) = \chi_{h^{-1}}(g)s_e,$$

showing that $\alpha_h(U(g)) = \chi_{h^{-1}}(g)U(g)$. To show that $\chi_h(g) = \chi_g(h)$, we first note that

$$p(U(g^{-1}))s_e \in \mathrm{Hom}_{\mathcal{D}}(1, \rho^2 \circ \alpha_{g^{-1}}) = \mathrm{Hom}_{\mathcal{D}}(\alpha_g, \rho^2).$$

In other words, $\rho(U(g^{-1}))s_e = cs_g$ for some $c \in \Bbbk$. To show that $c = 1$, we see that

$$d_\pm s_e'\rho(\rho(U(g^{-1}))s_e) = d_\pm s_e'\rho^2(U(g^{-1}))\rho(s_e) = d_\pm U(g^{-1})s_e'\rho(s_e) = U(g^{-1}),$$

while

$$d_\pm s_e'\rho(cs_g) = cd_\pm s_e'U(g)\rho(s_e)U(g)^{-1} = cd_\pm s_e'\rho(s_e)U(g^{-1}) = cU(g^{-1}).$$

It follows that

$$d_\pm\rho(\alpha_h(s_e'))\rho(\alpha_h(U(g)))\rho(U(g^{-1}))s_e = d_\pm\rho(\alpha_h(s_e'))\rho(\alpha_h(U(g)))s_g.$$

But the left-hand side gives us

$$\begin{aligned}
d_\pm\rho(\alpha_h(s_e'))\rho(\alpha_h(U(g)))\rho(U(g^{-1}))s_e &= \chi_{h^{-1}}(g)d_\pm\rho(\alpha_h(s_e'))\rho(U(g))\rho(U(g^{-1}))s_e \\
&= \chi_{h^{-1}}(g)d_\pm\rho(\alpha_h(s_e'))s_e \\
&= \chi_{h^{-1}}(g)d_\pm U(h)\rho(s_e')U(h)^{-1}s_e \\
&= \chi_{h^{-1}}(g)U(h),
\end{aligned}$$

while the right-hand side gives us

$$\begin{aligned}
d_\pm\rho(\alpha_h(s_e'))\rho(\alpha_h(U(g)))s_g &= d_\pm\rho(\alpha_h(s_e'U(g)))s_g \\
&= d_\pm\rho(\alpha_h(s_e'))s_g \\
&= d_\pm U(h)\rho(s_e')U(h)^{-1}s_g \\
&= \chi_g(h^{-1})d_\pm U(h)\alpha_g(\rho(s_e')s_e) \\
&= \chi_g(h^{-1})U(h).
\end{aligned}$$

The result follows. This completes the proof. ■

What have we actually done?

(i). two finite-dimensional vector spaces $\mathcal{K}$ and $\mathcal{K}'$;
(ii). four linear maps $j_1, j_2 \in \mathrm{Hom}(\mathcal{K}, \mathcal{K}')$ and $j_1', j_2' \in \mathrm{Hom}(\mathcal{K}', \mathcal{K})$;
(iii). two representations $U_{\mathcal{K}}, V : G \to \mathrm{Aut}(\mathcal{K})$;
(iv). a symmetric bicharacter $\chi : G \times G \to \Bbbk^\times$;
(v). two linear maps $l \in \mathrm{Hom}(\mathcal{K}, \mathcal{K} \otimes \mathcal{K})$ and $l' \in \mathrm{Hom}(\mathcal{K}', \mathcal{K}' \otimes \mathcal{K}')$.

We have not made any choices, except for rescaling.

Recall that the Leavitt algebra $\mathcal{L}$ admits an involutive antiautomorphism $-' : \mathcal{L} \to \mathcal{L}$ that exchanges each s_g and t_i with their primed versions and vice versa. Note that ρ, α_g and φ are NOT? compatible with this map; $\rho(-)' \neq \rho(-')$, $\alpha_g(-)' \neq \alpha_g(-')$ and $\varphi(-)' \neq \varphi'(-')$. We shall henceforth define φ and l on $\mathcal{K}'$ via φ' and l', respectively. Double-check this stuff. Maybe replace mentions of φ' and l' in what follows. Check Corollary 1 and Lemma 2 of EG17, i.e. $\mathrm{Hom}(\beta, \gamma) = \mathrm{Hom}(\tilde{\beta}, \tilde{\gamma})$.

Our setting diverges from the unitary setting here due to the fact that $U(g)'$ is not necessarily equal to $U(g)^{-1}$, since $\overline{\chi_h(g)}$ is not necessarily equal to $\chi_h(g)^{-1}$ and $U_{\mathcal{K}}(g)'$ is not necessarily equal to $U_{\mathcal{K}}(g)^{-1}$.

TO-DO: possibly revisit the other conditions in Izumi's lemma 3.4.

Prove other properties. The following results follow the work of Izumi closely. First, we work up to a classification theorem.

In what follows, we will let λ denote the left regular representation.

Theorem 4.9. *If $d_\pm$ is irrational, then*

(i). *G is necessarily Abelian;*
(ii). *m is a multiple of n;*
(iii). *$\bigoplus_{h\in G} \chi_h \cong \lambda$;*
(iv). *$V \cong U_{\mathcal{K}} \cong \frac{m}{n}\lambda$.*

Conversely, if $d_\pm$ is rational, there exist two natural numbers s and t such that $d_\pm = st$, $m = (s-1)t$ and $n = st^2$. In particular,

(i). *there exist two natural numbers s and t such that $d_\pm = st$, $m = (s-1)t$ and $n = st^2$;*
(ii). *when $t = 1$,*
  - *G is Abelian,*
  - *χ_h is trivial for all $h \in G$,*
  - *$1 \oplus V \cong 1 \oplus U_{\mathcal{K}} \cong \lambda$;*

(iii). *when $t > 1$,*
  - *G is non-Abelian,*
  - *$|\mathrm{Hom}(G, \Bbbk^\times)| = t^2$,*
  - *$\oplus_{h\in G}\chi_h \cong \oplus_{\chi\in \mathrm{Hom}(G,\Bbbk^\times)}\chi^{\oplus s}$,*
  - *t divides $\dim(\pi)$, for all $\pi \in \hat{G}^\dagger$, where $\hat{G}^\dagger$ is the set of equivalence classes of irreducible unitary representations of G with dimension greater than 1,*
  - *$V \cong U_{\mathcal{K}} \cong \bigoplus_{\pi\in\hat{G}^\dagger} \frac{\dim(\pi)}{t}\pi$.*

Proof. For all $g \in G$, we have $\delta_{g,e} = \rho(s_e' s_g) = \rho(s_e')U(g)\rho(s_e)U(g)^{-1}$. Since $U(e) = 1$,

$$\delta_{g,e} = \rho(s_e')U(g)\rho(s_e) = \frac{1}{d_\pm^2}\sum_{h\in G}\chi_h(g) + \sum_{i,j=1}^{m}\rho(s_e')t_i t_i' U_{\mathcal{K}}(g) t_j t_j' \rho(s_e).$$

Observe that $t_i' U_{\mathcal{K}}(g) t_j \in \Bbbk$, as it is in the 1-dimensional space $\mathrm{Hom}_{\mathcal{D}}(\rho, \rho)$. We therefore have

$$\begin{aligned}\delta_{g,e} &= \frac{1}{d_\pm^2} \sum_{h \in G} \chi_h(g) + \sum_{i,j=1}^{m} \rho(s_e') t_i t_i' U_{\mathcal{K}}(g) t_j t_j' \rho(s_e) \\ &= \frac{1}{d_\pm^2} \sum_{h \in G} \chi_h(g) + \frac{1}{d_\pm} \sum_{i=1}^{m} t_i' U_{\mathcal{K}}(g) t_i \\ &= \frac{1}{d_\pm^2} \sum_{h \in G} \chi_h(g) + \frac{1}{d_\pm} \mathrm{Tr}(U_{\mathcal{K}}(g)).\end{aligned}$$

Because $\mathrm{Tr}(\lambda(g)) = n\delta_{g,e}$, we have

$$\mathrm{Tr}(\lambda(g)) = \frac{n}{d_\pm^2} \sum_{h \in G} \chi_h(g) + \frac{n}{d_\pm} \mathrm{Tr}(U_{\mathcal{K}}(g)).$$

Let $\chi \in \mathrm{Hom}(G, \Bbbk^\times)$ now be a 1-dimensional representation. Denote by a_χ and b_χ the multiplicities of χ in the representations $\bigoplus_{h \in G} \chi_h$ and $U_{\mathcal{K}}$, respectively. Of course, χ has multiplicity 1 in λ. Because the character of a direct sum of representations is the direct sum of the characters of those representations, inspecting the coefficient of χ on both sides of the previous equation gives us

$$d_\pm^2 = n a_\chi + n d_\pm b_\chi.$$

If $d_\pm$ is irrational, we have no choice but to take $a_\chi = 1$ and $n b_\chi = m$. The former implies that

$$\bigoplus_{h \in G} \chi_h \cong \bigoplus_{\chi \in \mathrm{Hom}(G, \Bbbk^\times)} \chi.$$

In other words, $|G| = |\mathrm{Hom}(G, \Bbbk^\times)|$, whence G must be Abelian.

Conversely, suppose that $d_\pm$ is rational. Then one can show that the integer $m \pm \sqrt{m^2 + 4n}$ is even and hence that $d_\pm$ is an integer. Now, any object tensored with its dual is Morita equivalent to 1, so $\rho \otimes \rho$ is an algebra object with dimension $d_\pm^2$. The restriction of $\rho \otimes \rho$ to Vec_G, given by throwing away simple summands not in Vec_G, is $\bigoplus_{g \in G} \alpha_g$ with dimension n. Thus by Theorem 8.4, there exists an algebra object with dimension $d_\pm^2/n$. Because this must be a sum of simple objects with dimensions $d_\pm$ or 1, it follows that $d_\pm^2/n = 1 + m d_\pm/n$ must be an integer. Let $s := 1 + m d_\pm/n$ and $t := m/(s-1)$. Note that $d_\pm^2 = n + md$ together with $n(s-1) = md$ implies $n(s-1)^2 = sm^2$, whence t must also be an integer. In particular, $d_\pm = st$ and $n = st^2$. This is pretty tedious. I'll finish it later. Need to adapt the category theory parts, since we no longer have a category. ■

Does this theorem imply that χ is non-degenerate (that is, $\chi_h(g) = 1$ for all $h \in G$ if and only if $g = e$)?

Lemma 4.10. *We have*

$$\rho(U(g)) = \sum_{h\in G} s_h s'_{hg} + \varphi U_{\mathcal{K}}(g)\varphi' \otimes U(g).$$

for all $g \in G$.

Proof. We have shown in the proof for Corollary 4.8 that $\rho(U(g))s_h = s_{hg^{-1}}$. Thus

$$\rho(U(g))S = \sum_{h\in G} \rho(U(g))s_h s'_h = \sum_{h\in G} s_{hg^{-1}} s'_h.$$

It follows that both S and T commute with $\rho(U(g))$, since

$$\rho(U(g))T = \rho(U(g))(S+T) - \rho(U(g))S = (S+T)\rho(U(g)) - S\rho(U(g)) = T\rho(U(g)).$$

Thus $\rho(U(g))T = \rho(U(g))T^2 = T\rho(U(g))T$ and

$$\rho(U(g))(S+T) = \sum_{h\in G} s_h s'_{hg} + T\rho(U(g))T.$$

Now, $t'_i\rho(U(g))t_j \in \mathrm{Hom}_{\mathsf{End}(A)}(\rho, \rho\otimes\alpha_g) = \Bbbk U(g)$. That is, $t'_i\rho(U(g))t_j = U'(g)_{ij}U(g)$ for some $U'(g)_{ij} \in \Bbbk$. Thus $T\rho(U(g))T = U'(g)\otimes U(g)$, where the matrix $U'(g)$ is determined by

$$s'_e t'_i \rho(U(g)) t_j s_e = U'(g)_{ij} = t'_i U'(g) t_j.$$

Substituting

$$U(g) = \sum_{h\in G} \chi_h(g) s_h s'_h + \sum_{p,q=1}^{m} U_{\mathcal{K}}(g)_{pq} t_p t'_q,$$

we obtain

$$\begin{aligned} s'_e t'_i \rho(U(g)) t_j s_e &= \sum_{h\in G} \chi_h(g) s'_e t'_i U(h)\rho(s_e s'_e)U(h)^{-1} t_j s_e + \sum_{p,q=1}^{m} U_{\mathcal{K}}(g)_{pq} s'_e t'_i \rho(t_p t'_q) t_j s_e \\ &= \sum_{p,q=1}^{m} U_{\mathcal{K}}(g)_{pq} s'_e t'_i \rho(t_p t'_q) t_j s_e. \end{aligned}$$

Because $t'_i\rho(t_p)$ and $\rho(t'_q)t_j$ are in $\mathrm{Hom}_{\mathsf{End}(A)}(\rho\otimes\rho, \rho\otimes\rho)$, they are spanned by $\{s_g s'_g\}_{g\in G} \sqcup \{t_i t'_j\}_{i,j=1}^{m}$, just like $U(g)$. We therefore have that

$$t'_i\rho(t_p) = \sum_{h\in G} a_h(g) s_h s'_h + \sum_{k,l=1}^{m} a_{kl}(g) t_k t'_l,$$

$$\rho(t_q)t_j = \sum_{h\in G} b_h(g) s_h s'_h + \sum_{k,l=1}^{m} b_{kl}(g) t_k t'_l$$

and hence

$$\begin{aligned}U'(g)_{ij} = s'_e t'_i \rho(U(g)) t_j s_e &= \sum_{p,q=1}^{m} U_{\mathcal{K}}(g)_{pq} s'_e t'_i \rho(t_p)\rho(t'_q) t_j s_e = \sum_{p,q=1}^{m} U_{\mathcal{K}}(g)_{pq} a_e(g) b_e(g)\\ &= \sum_{p,q=1}^{m} U_{\mathcal{K}}(g)_{pq} d^2 \rho(s'_e) t'_i \rho(t_p) s_e s'_e \rho(t'_q) t_j \rho(s_e)\\ &= \sum_{p,q=1}^{m} U_{\mathcal{K}}(g)_{pq} d^2 t'_i \rho^2(s'_e) \rho(t_p) s_e s'_e \rho(t'_q) \rho^2(s_e) t_j\\ &= \sum_{p,q=1}^{m} U_{\mathcal{K}}(g)_{pq} t'_i \varphi(t_p) \varphi'(t'_q) t_j.\end{aligned}$$

Thus

$$U'(g) = \sum_{i,j=1}^{m} t_i U'(g)_{ij} t'_j = \sum_{p,q=1}^{m} T\varphi(t_p) U_{\mathcal{K}}(g)_{pq} \varphi'(t'_q) T = \sum_{p,q=1}^{m} \varphi t_p U_{\mathcal{K}}(g)_{pq} t'_q \varphi' = \varphi U_{\mathcal{K}}(g) \varphi',$$

whence the result follows. This completes the proof. ■

This is the first point where our setting diverges from the unitary setting.

Lemma 4.11. *Let $\pi : G \to \mathrm{End}(A)$ and $\sigma : G \to \mathrm{End}(B)$ be irreducible representations of G. Then*

$$\sum_{g\in G} \pi(g) \circ \gamma \circ \sigma(g)^{-1} = n\delta_{\pi,\sigma}$$

for any $\gamma : B \to A$.

Proof. Define

$$\psi := \sum_{g\in G} \pi(g) \circ \gamma \circ \sigma(g)^{-1}.$$

Then for any $h \in G$, we have that

$$\pi(h) \circ \psi \circ \sigma(h)^{-1} = \sum_{g\in G} \pi(hg) \circ \gamma \circ \sigma(hg)^{-1} = \psi.$$

Thus ψ intertwines π and σ. Provided $\Bbbk$ is algebraically closed, it follows from Schur's lemma that

$$\sum_{g\in G} \pi(g) \circ \gamma \circ \sigma(g)^{-1} = n\delta_{\pi,\sigma}.$$

This completes the proof. ■

Lemma 4.12. *Let $\hat{G}^V$ be the set of irreducible summands of V and let*

$$(U_{\mathcal{K}}, \mathcal{K}) = \bigoplus_{\pi\in\hat{G}^V} \bigoplus_{a=1}^{m_\pi} (\pi, \mathcal{K}^a_\pi)$$

be the corresponding internal direct sum decomposition, where m_π is the multiplicity of π. Moreover, let $\{t^a_{\pi,i}\}_{i=1}^{\dim(\pi)}$ be a basis for $\mathcal{K}^a_\pi \subseteq \mathcal{K}$ such that

$$U_{\mathcal{K}}(g) = \sum_{\pi\in\hat{G}^V}\sum_{a=1}^{m_\pi} \pi(g) t^a_\pi {t^a_\pi}' = \sum_{\pi\in\hat{G}^V}\sum_{a=1}^{m_\pi}\sum_{i,j=1}^{\dim(\pi)} \pi(g)_{ij} t^a_{\pi,i} {t^a_{\pi,j}}',$$

where $\{{t^a_{\pi,i}}'\}_{i=1}^{\dim(\pi)}$ is the unique set of projections corresponding to $\{t^a_{\pi,i}\}_{i=1}^{\dim(\pi)}$ by [Mac13, Theorem VIII.2.2]. Then

$$\sum_{\pi,a,b,i,j} \frac{n}{\dim(\pi)} t^a_{\pi,i} {t^a_{\pi,j}}' \rho(s_e)\rho(s_e') t^b_{\pi,j} {t^b_{\pi,j}}' + \sum_{i=1}^m l(t_i) l'(t_i) = T\otimes T,$$

$$\sum_{i=1}^m l(t_i) V(h) \rho(t_i') s_e + \frac{\nu n}{\sqrt{d}} \sum_{\pi,a} \delta_{\chi_h,\pi} t^a_\pi {t^a_\pi}' \rho(s_e) = 0.$$

Proof. To begin, we first note that by Lemma **??**,

$$\begin{aligned}
\sum_{g\in G} \rho(s_g)\rho(s_g') &= \sum_{g\in G} U(g)\rho(s_e)\rho(s_e')U(g)^{-1} \\
&= \frac{1}{d^2}\sum_{g,h,k\in G} \chi_h(g)\chi_k(g)^{-1} s_h s_k' + \\
&\quad \frac{\nu}{d}\sum_{g,k\in G}\sum_{\pi,a,i,j} \pi(g)_{ij}\chi_k(g)^{-1} t^a_{\pi,i}{t^a_{\pi,j}}'\rho(s_e)s_k' + \\
&\quad \frac{\nu}{d}\sum_{g,h\in G}\sum_{\pi,a,i,j} \chi_h(g)\pi(g)^{-1}_{ij} s_h \rho(s_e') t^a_{\pi,i}{t^a_{\pi,j}}' + \\
&\quad \sum_{g\in G}\sum_{\pi,a,i,j}\sum_{\sigma,b,p,q} \pi(g)_{ij}\sigma(g)^{-1}_{pq} t^a_{\pi,i}{t^a_{\pi,j}}'\rho(s_e)\rho(s_e') t^b_{\sigma,p}{t^b_{\sigma,q}}'.
\end{aligned}$$

By Lemma 4.11, check $1/\dim(\pi)$

$$\begin{aligned}
\sum_{g\in G} \rho(s_g)\rho(s_g') &= \frac{n}{d^2}\sum_{h,k\in G} \delta_{\chi_h,\chi_k} s_h s_k' + \\
&\quad \frac{n\nu}{d}\sum_{k\in G}\sum_{\pi,a} \delta_{\pi,\chi_k} t^a_\pi {t^a_\pi}'\rho(s_e)s_k' + \\
&\quad \frac{n\nu}{d}\sum_{h\in G}\sum_{\pi,a} \delta_{\chi_h,\pi} s_h\rho(s_e') t^a_\pi {t^a_\pi}' + \\
&\quad n\sum_{g\in G}\sum_{\pi,a,i,j} \frac{1}{\dim(\pi)} t^a_{\pi,i}{t^a_{\pi,j}}'\rho(s_e)\rho(s_e') t^b_{\pi,i}{t^b_{\pi,j}}'.
\end{aligned}$$

Conversely, by Lemma **??**,

$$\sum_{i=1}^m \rho(t_i)\rho(t_i') = \sum_{i=1}^m \sum_{h,k\in G} s_h s_h' \rho(t_i)\rho(t_i') s_k s_k' + \sum_{i=1}^m \sum_{h,k\in G} s_h s_h' \rho(t_i) s_k s_k' \varphi'(t_i') V(k)^{-1} +$$
$$\sum_{i=1}^m \sum_{h\in G} s_h s_h' \rho(t_i) l'(t_i') + \sum_{i=1}^m \sum_{h,k\in G} V(h)\varphi(t_i) s_h s_h' \rho(t_i') s_k s_k' +$$
$$\sum_{i=1}^m \sum_{h\in G} V(h)\varphi(t_i) s_h s_h' \varphi'(t_i') V(h)^{-1} + \sum_{i=1}^m \sum_{h\in G} V(h)\varphi(t_i) s_h s_h' l'(t_i') +$$
$$\sum_{i=1}^m \sum_{h\in G} l(t_i)\rho(t_i') s_h s_h' + \sum_{i=1}^m \sum_{h\in G} l(t_i) s_h s_h' \varphi'(t_i') V(h)^{-1} + \sum_{i=1}^m l(t_i) l'(t_i').$$

Now, note that $\rho(t_i')s_k \in \mathcal{K}$, while $s_h'\rho(t_i) \in \mathcal{K}'$. In particular, we have check this

$$\sum_{i=1}^m s_h' \rho(t_i)\rho(t_i') s_k = \sum_{i=1}^m s_e' \rho(t_i) V(h^{-1}k)\rho(t_i') s_e = \mathrm{Tr}(V(h^{-1}k)).$$

Somehow $V(h)\varphi(t_i) = t_i$? Since V is equivalent to $U_{\mathcal{K}}$, it follows from Theorem 4.9 that

$$\mathrm{Tr}(V(h^{-1}k)) = \mathrm{Tr}(U_{\mathcal{K}}(h^{-1}k)) = \frac{d}{n}\mathrm{Tr}(\lambda(h^{-1}k)) - \frac{1}{d}\sum_{g\in G} \chi_k(g)\chi_h(g)^{-1} = d\delta_{h,k} - \frac{n}{d}\delta_{\chi_h,\chi_k}.$$

Conclude using $S + T = 1$. This completes the proof. ∎

Lemma 4.13. *In terms of l and l', Lemma 4.10 is equivalent to*

$$\varphi U_{\mathcal{K}}(g)\varphi' \otimes U_{\mathcal{K}}(g) = \sum_{\pi,\sigma,a,b,i,j,p,q} \sum_{h\in G} \chi_g(h)\pi(h)_{ij}\sigma(h)^{-1}_{pq} t^a_{\pi,i} {t^a_{\pi,j}}' \rho(s_e)\rho(s_e') t^b_{\sigma,q} {t^b_{\sigma,p}}' +$$
$$\sum_{\pi,a,i,j} \pi(g)_{ij} l(t^a_{\pi,i}) l'({t^a_{\pi,j}}').$$

Proof. Note that

$$S\varphi U_{\mathcal{K}}(g)\varphi' = 0.$$

In other words, Lemma 4.10 holds if and only if

$$S\rho(U(g))S = \sum_{h\in G} s_h s_{hg}' \quad \text{and} \quad T\rho(U(g))T = \varphi U_{\mathcal{K}}(g)\varphi' \otimes U(g).$$

Now, $S\rho(U(g))S$ does not give any condition on l, since

.

On the other hand, for $T\rho(U(g))T$, we have

$$T\rho(U(g))T = \sum_{h\in G} \chi_h(g) T U(h)\rho(s_e)\rho(s_e') U(h)^{-1} T + \sum_{\pi,a,i,j} \pi(g)_{ij} T \rho(t^a_{\pi,i} {t^a_{\pi,j}}') T.$$

The first term is simply

$$\sum_{h\in G}\chi_h(g)TU(h)\rho(s_es_e')U(h)^{-1}T=\sum_{\pi,\sigma,a,b,i,j,p,q}\sum_{h\in G}\chi_g(h)\pi(h)_{ij}\sigma(h)^{-1}_{pq}t^a_{\pi,i}{t^a_{\pi,j}}'\rho(s_es_e')t^b_{\sigma,q}{t^b_{\sigma,p}}'.$$

As for the second, check this, period 3 condition?

$$\begin{aligned}\sum_{\pi,a,i,j}\pi(g)_{ij}T\rho(t^a_{\pi,i}{t^a_{\pi,j}}')T &= \sum_{h\in G}\sum_{\pi,a,i,j}\pi(g)_{ij}V(h)\varphi(t^a_{\pi,i})s_hs_h'\varphi({t^a_{\pi,j}}')V(h)^{-1}+\sum_{\pi,a,i,j}\pi(g)_{ij}l(t^a_{\pi,i})l'({t^a_{\pi,j}}')\\ &= \sum_{h\in G}V(h)\varphi U_{\mathcal{K}}(g)\varphi'V(h)^{-1}\otimes s_hs_h'+\sum_{\pi,a,i,j}\pi(g)_{ij}l(t^a_{\pi,i})l'({t^a_{\pi,j}}')\\ &= \sum_{h\in G}\varphi U_{\mathcal{K}}(g)\varphi'\otimes\chi_h(g)s_hs_h'+\sum_{\pi,a,i,j}\pi(g)_{ij}l(t^a_{\pi,i})l'({t^a_{\pi,j}}').\end{aligned}$$

The result follows, since

$$\varphi U_{\mathcal{K}}(g)\varphi'\otimes U(g)-\sum_{h\in G}\varphi U_{\mathcal{K}}(g)\varphi'\otimes\chi_h(g)s_hs_h'=\varphi U_{\mathcal{K}}(g)\varphi'\otimes U_{\mathcal{K}}(g).$$

This completes the proof. ■

Check Corollary 1 and Lemma 2 of EG17, i.e. $\mathrm{Hom}(\beta,\gamma)=\mathrm{Hom}(\tilde\beta,\tilde\gamma)$.

5. General Reconstruction

In

6. Irrational Reconstruction

Let $v_0 \coloneqq V$, $v_1 \coloneqq U_K$, $v_2 \coloneqq \varphi U_K(g)\varphi^{-1}$. Then by Lemma 4.6 and Lemma 4.7, these satisfy

$$\varphi v_i(g)\varphi^{-1} = v_{i+1}(g) \qquad \text{and} \qquad v_{i+1}(g)v_i(h) = \chi_h(g)v_i(h)v_{i+1}(g),$$

for $i \in \mathbb{Z}/3\mathbb{Z}$. This motivates the following lemma.

Lemma 6.1.

7. Reconstruction for $m = |G|$

Let G be a finite Abelian group of order n and $a : G \to \Bbbk^\times$ a self-inverse projective representation whose 2-cocycle is given by a non-degenerate symmetric bicharacter $\langle \cdot, \cdot \rangle : G \times G \to \Bbbk^\times$. That is,

$$a(e) = 1, \qquad a(g)a(h) = \langle g, h \rangle a(gh) \qquad \text{and} \qquad a(g) = a(g^{-1}),$$

for all $g, h \in G$. Moreover, suppose we have $b : G \to \Bbbk$ and $c \in \Bbbk^\times$ such that

$$\begin{aligned} a(g)b(g^{-1}) &= b(g)^{-1}, \\ c\frac{\sqrt{n}}{d} + \sum_{g \in G} b(g) &= 0, \\ \sum_{g \in G} b(gh)b(g)^{-1} &= \delta_{h,0} - \frac{1}{d}, \end{aligned}$$

where $d^2 = nd + n$. Let $\mathcal{L}_{2n}$ be the Leavitt algebra with generators $\{s_g, t_h\}_{g,h \in G} \sqcup \{s'_g, t'_h\}_{g,h \in G}$. We define an action $\alpha : G \curvearrowright \mathcal{L}_{2n}$ and a representation $U : G \to \mathcal{L}_{2n}$ by

$$\begin{aligned} \alpha_g(s_h) &:= s_{gh}, \qquad \alpha_g(t_h) := \langle g, h \rangle t_h, \\ U(g) &:= \sum_{h \in G} \langle g, h \rangle s_h s'_h + \sum_{h \in G} t_{hg^{-1}} t'_h. \end{aligned}$$

What do we need to show?

Determine how ρ and α_g act on the generators!

For each set of fusion rules, find solution set of endomorphisms ρ and α_g. Then for each possible solution, show that $\overline{\mathsf{End}(\mathcal{A})}_{\oplus}$ is a fusion category.

Temporary results.

Let $\gamma(g) := \pi(g)\sigma(g)^{-1}$. If $\gamma(g) = 1$ for all $g \in G$, then

$$\sum_{g\in G} \pi(g)\sigma(g)^{-1} = \sum_{g\in G} \gamma(g) = n,$$

and moreover $\gamma(g) = 1$ if and only if $\pi = \sigma$. Suppose instead that there exists some $x \in G$ for which $\gamma(x) \neq 1$. Then

$$\gamma(x) \sum_{g\in G} \gamma(g) = \sum_{g\in G} \gamma(xg) = \sum_{g\in G} \gamma_{h,k}(g),$$

whence

$$(\gamma_{h,k}(x) - 1) \sum_{g\in G} \gamma_{h,k}(g) = 0.$$

Since $\gamma_{h,k}(x) \neq 1$, and assuming that $\gamma_{h,k}(x) - 1$ is invertible in $\Bbbk$, the sum must be zero. Thus

$$\sum_{g\in G} \chi_h(g)\chi_k(g)^{-1} = n\delta_{\chi_h,\chi_k}.$$

Part 2:
Let $\pi : G \to \mathrm{GL}(V)$ and $\sigma : G \to \mathrm{GL}(W)$ be irreducible representations of a finite group G. Define

$$\psi := \sum_{g\in G} \pi(g)\sigma(g)^{-1}.$$

Then for any $h \in G$, we have that

$$\pi(h)\psi\sigma(h)^{-1} = \sum_{g\in G} \pi(hg)\sigma(hg)^{-1} = \psi.$$

In other words, ψ intertwines π and σ. Provided $\Bbbk$ is algebraically closed, Schur's lemma implies that $\psi = \lambda\delta_{\pi,\sigma}$ for some $\lambda \in \Bbbk$. Finally, since $n = \mathrm{Tr}(\psi) = \lambda \dim(\pi)$ (by prev result), it follows that

$$\sum_{g\in G} \pi(g)\sigma(g)^{-1} = \frac{n}{\dim(\pi)}\delta_{\pi,\sigma}.$$

8. Appendix

Proposition 8.1. *Let $\mathcal{C}$ be a $\Bbbk$-linear fusion category with group G of invertible objects, and suppose there exists a simple object $X \in \mathrm{Ob}(\mathcal{C})$ such that $g \otimes X \cong X$ for all $g \in G$. Then the full Abelian monoidal subcategory generated by G is monoidally equivalent to Vec_G with trivial associativity.*

Proof. Let $\mathcal{D}$ be the subcategory generated by G. This category is easily seen to be fusion, and is hence equivalent to Vec_G on the level of fusion rings. It thus suffices to show that the associativity is trivial. We first choose isomorphisms $f_g \in \mathrm{Hom}_{\mathcal{C}}(g \otimes X, X)$ and $\mu'_{g,h} \in \mathrm{Hom}_{\mathcal{C}}(g \otimes h, gh)$ for all $g, h \in G$. Because $\mathrm{Hom}_{\mathcal{C}}((g \otimes h) \otimes X, gh \otimes X)$ is 1-dimensional, there exists some $z_{g,h} \in \Bbbk^\times$ for which

$$z_{g,h}\mu'_{g,h} \otimes \mathrm{id}_X = z_{g,h}(\mu'_{g,h} \otimes \mathrm{id}_X) = f_{gh}^{-1} \circ f_g \circ (\mathrm{id}_g \otimes f_h) \circ \alpha_{g,h,X}.$$

That is, for each pair $g, h \in G$, there is a unique isomorphism $\mu_{g,h} \coloneqq z_{g,h}\mu'_{g,h}$ for which the diagram

$$\begin{tikzcd}
(g \otimes h) \otimes X \arrow[r, "\mu_{g,h}\otimes \mathrm{id}_X"] \arrow[dd, "\alpha_{g,h,X}"'] & gh \otimes X \arrow[dr, "f_{gh}"] & \\
& & X \\
g \otimes (h \otimes X) \arrow[r, "\mathrm{id}_g \otimes f_h"'] & g \otimes X \arrow[ur, "f_g"'] &
\end{tikzcd}$$

commutes. From this, we obtain the commutative diagram

$$\begin{tikzcd}
((g\otimes h)\otimes k)\otimes X \arrow[rrrr, "(\mu_{g,h}\otimes \mathrm{id}_k)\otimes \mathrm{id}_X"] \arrow[dr, "\alpha_{g\otimes h,k,X}"] \arrow[dddddd, "\alpha_{g,h,k}\otimes \mathrm{id}_X"'] & & & & (gh\otimes k)\otimes X \arrow[dl, "\alpha_{gh,k,X}"'] \arrow[dddr, "\mu_{gh,k}\otimes \mathrm{id}_X"] & \\
& (g\otimes h)\otimes(k\otimes X) \arrow[rr, "\mu_{g,h}\otimes \mathrm{id}_{k\otimes X}"] \arrow[dr, "\mathrm{id}_{g\otimes h}\otimes f_k"] \arrow[dd, "\alpha_{g,h,k\otimes X}"] & & gh\otimes(k\otimes X) \arrow[d, "\mathrm{id}_{gh}\otimes f_k"'] & & \\
& & (g\otimes h)\otimes X \arrow[r, "\mu_{g,h}\otimes \mathrm{id}_X"'] \arrow[dd, "\alpha_{g,h,X}"] & gh\otimes X \arrow[dr, "f_{gh}"'] & & \\
& g\otimes(h\otimes(k\otimes X)) \arrow[dr, "\mathrm{id}_g\otimes(\mathrm{id}_h\otimes f_k)"] & & & X & ghk\otimes X \arrow[l, "f_{ghk}"'] \, . \\
& & g\otimes(h\otimes X) \arrow[r, "\mathrm{id}_g\otimes f_h"] & g\otimes X \arrow[ur, "f_g"] & & \\
& g\otimes((h\otimes k)\otimes X) \arrow[uu, "\mathrm{id}_g\otimes\alpha_{h,k,X}"'] \arrow[rr, "\mathrm{id}_g\otimes(\mu_{h,k}\otimes \mathrm{id}_X)"'] & & g\otimes(hk\otimes X) \arrow[u, "\mathrm{id}_g\otimes f_{hk}"] & & \\
(g\otimes(h\otimes k))\otimes X \arrow[ur, "\alpha_{g,h\otimes k,X}"'] \arrow[rrrr, "(\mathrm{id}_g\otimes\mu_{h,k})\otimes \mathrm{id}_X"'] & & & & (g\otimes hk)\otimes X \arrow[ul, "\alpha_{g,hk,X}"] \arrow[uuur, "\mu_{g,hk}\otimes \mathrm{id}_X"'] &
\end{tikzcd}$$

Note that the leftmost pentagon is the pentagon diagram for the associator, while all other pentagons are versions of the pentagon given at the beginning of the proof. The upper middle quadrilateral comes from the functoriality of $\otimes$, with the remaining quadrilaterals come from the naturality of α. If we look at the outermost pentagon, we obtain the commutative diagram

$$\begin{tikzcd}
(g \otimes h) \otimes k \arrow[r, "\mu_{g,h} \otimes \mathrm{id}_k"] \arrow[dd, "\alpha_{g,h,k}"'] & gh \otimes k \arrow[dr, "\mu_{gh,k}"] & \\
& & ghk \\
g \otimes (h \otimes k) \arrow[r, "\mathrm{id}_g \otimes \mu_{h,k}"'] & g \otimes hk \arrow[ur, "\mu_{g,hk}"'] &
\end{tikzcd} \cdot$$

Thus the associator for $\mathcal{D}$ is given by

$$\alpha_{g,h,k} = (\mathrm{id}_g \otimes \mu_{h,k}^{-1}) \circ \mu_{g,hk}^{-1} \circ \mu_{gh,k} \circ (\mu_{g,h} \otimes \mathrm{id}_k).$$

We claim that α is cohomologous to the trivial 3-cocycle. Indeed, let $\alpha^1_{g,h,k}$ denote the components of the trivial associator in $\mathcal{D}$, for $g, h, k \in G$. This morphism factors as

$$\alpha^1_{g,h,k} = (\mathrm{id}_g \otimes \beta_{h,k}^{-1}) \circ \beta_{g,hk}^{-1} \circ \beta_{gh,k} \circ (\beta_{g,h} \otimes \mathrm{id}_k),$$

where we define $\beta_{g,h} \in \mathrm{Hom}_{\mathcal{C}}(g \otimes h, gh)$ by $\beta_{g,h}(u \otimes v) \coloneqq uv$ for all $g, h \in G$. By linearity, there exists a 3-cochain $a : G^2 \to \Bbbk^\times$ such that $\alpha_{g,h,k} = a(g,h,k)\alpha^1_{g,h,k}$. Moreover, for all $g, h \in G$ there exists a 2-cochain $b : G^2 \to \Bbbk^\times$ satisfying $\mu_{g,h} = b(g,h)\beta_{g,h}$. It follows that

$$a(g,h,k) = b(g,h)^{-1} b(gh,k) b(g,hk)^{-1} b(g,h).$$

In particular, it is clear that $a^{-1} = d^2(b)$, where d^2 is the second differential, implying that a^{-1} and hence a are 3-coboundaries. The claim follows, with the classification of monoidal structures on Vec_G ([EGNO16, Proposition 2.6.1]) showing that $\mathcal{D}$ is monoidally equivalent to Vec_G with trivial associativity. This completes the proof. ■

Lemma 8.2. *Let $\mathcal{C}$ be a monoidal category with rigid object X isomorphic to its dual. Then there exists an evaluator and a coevaluator such that the dual of X is exactly equal to X.*

Proof. We will consider the left duals, but note that the following proof holds similarly for right duals. Let $f_X : X \to X^*$ be an isomorphism from X to its left dual. We define a new evaluator and a new coevaluator by

$$\overline{\mathrm{ev}}_X \coloneqq \mathrm{ev}_X \circ (f_X \otimes \mathrm{id}_X) \qquad \text{and} \qquad \overline{\mathrm{coev}}_X \coloneqq (\mathrm{id}_X \otimes f_X^{-1}) \circ \mathrm{coev}_X,$$

respectively. This gives X the structure of a left dual to itself. Moreover, the left dual $f_X^* : X^{**} \to X^*$ allows us to give X^* the structure of a left dual to itself by defining

$$\overline{\mathrm{ev}}_{X^*} \coloneqq \mathrm{ev}_{X^*} \circ ({f_X^*}^{-1} \otimes \mathrm{id}_{X^*}) \qquad \text{and} \qquad \overline{\mathrm{coev}}_{X^*} \coloneqq (\mathrm{id}_{X^*} \otimes f_X^*) \circ \mathrm{coev}_{X^*}.$$

In particular, these are consistent in the sense that one can show (with some effort) that

$$\overline{\mathrm{ev}}_{X^*} \circ (f_X \otimes f_X) = \overline{\mathrm{ev}}_X \qquad \text{and} \qquad (f_X^{-1} \otimes f_X^{-1}) \circ \overline{\mathrm{coev}}_{X^*} = \overline{\mathrm{coev}}_X.$$

This completes the proof. ■

Remark 8.3. Let $\mathcal{C}$ be a $\Bbbk$-linear monoidal category with rigid object $X \cong X^*$. Furthermore, let $\mathrm{Hom}_{\mathcal{C}}(X, X^*) \cong \Bbbk$. Let's write down the left quantum trace

$$\mathrm{Tr}^L(\psi) := \mathrm{ev}_{X^*} \circ (\psi \otimes \mathrm{id}_{X^*}) \circ \mathrm{coev}_X$$

when $\psi \in \mathrm{Hom}_{\mathcal{C}}(X, X^{**})$ satisfies $\psi^* = \psi^{-1}$ and $f^{**} \circ \psi = \psi \circ f$ for all $f \in \mathrm{Hom}_{\mathcal{C}}(X, X^{**})$. Under Lemma 8.2, we can rewrite this in terms of the new evaluator and coevaluator as

$$\begin{aligned}\mathrm{Tr}^L(\psi) &= \overline{\mathrm{ev}}_{X^*} \circ (f_X^* \otimes \mathrm{id}_{X^*}) \circ (\psi \otimes \mathrm{id}_{X^*}) \circ (\mathrm{id}_X \otimes f_X) \circ \overline{\mathrm{coev}}_X \\ &= \overline{\mathrm{ev}}_X \circ (f_X^{-1} \otimes f_X^{-1}) \circ (f_X^* \otimes \mathrm{id}_{X^*}) \circ (\psi \otimes \mathrm{id}_{X^*}) \circ (\mathrm{id}_X \otimes f_X) \circ \overline{\mathrm{coev}}_X.\end{aligned}$$

Consider the map $r_X : f \mapsto f^* \circ \psi$ in $\mathrm{End}(\mathrm{Hom}_{\mathcal{C}}(X, X^*))$. Because $\mathrm{Hom}_{\mathcal{C}}(X, X^*)$ is 1-dimensional, $\mathrm{End}(\mathrm{Hom}_{\mathcal{C}}(X, X^*))$ is 1-dimensional with the canonical basis $\mathrm{id}_{\mathrm{Hom}_{\mathcal{C}}(X,X^*)} \mapsto \mathrm{id}_{\mathrm{Hom}_{\mathcal{C}}(X,X^*)}$. In other words, r_X is multiplication by a scalar. Since $r_X^2(f) = \psi^* \circ f^{**} \circ \psi = f$, it follows that $r_X(f) = \nu_\psi f$, for $\nu_\psi \in \{-1, 1\}$. This is precisely the (second) Frobenius–Schur indicator of X. We therefore have

$$\begin{aligned}\mathrm{Tr}^L(\psi) &= \overline{\mathrm{ev}}_X \circ (f_X^{-1} \otimes f_X^{-1}) \circ (f_X^* \otimes \mathrm{id}_{X^*}) \circ (\psi \otimes \mathrm{id}_{X^*}) \circ (\mathrm{id}_X \otimes f_X) \circ \overline{\mathrm{coev}}_X \\ &= \nu_\psi \, \overline{\mathrm{ev}}_X \circ (f_X^{-1} \otimes f_X^{-1}) \circ (f_X \otimes f_X) \circ \overline{\mathrm{coev}}_X \\ &= \nu_\psi \, \overline{\mathrm{ev}}_X \circ \overline{\mathrm{coev}}_X \\ &= \overline{\mathrm{ev}}_X \circ (\overline{\psi} \otimes \mathrm{id}_X) \circ \overline{\mathrm{coev}}_X.\end{aligned}$$

In a pivotal category, this tells us that a side effect of taking X to be self-dual is that the new pivotal structure $\overline{\psi}_X : X \to X$ becomes multiplication by the Frobenius–Schur indicator $\nu \in \{-1, 1\}$.

Theorem 8.4. *Let $\mathcal{C}$ be a fusion category with $\mathcal{D}$ a fusion subcategory, let A be an algebra object in $\mathcal{C}$ and let B be the restriction of A to $\mathcal{D}$, given by throwing away simple summands not in $\mathcal{D}$. Then*

(i). *B is an algebra object in $\mathcal{C}$;*
(ii). *B is a subalgebra object of A (in some sense);*
(iii). *A is a (B, A)-bimodule object by restriction;*
(iv). *there is a factoring of bimodule objects ${}_{\mathbb{1}}A_A = {}_{\mathbb{1}}B_B \otimes_B {}_BA_A$;*
(v). *the objects ${}_A(A^*)_B \otimes_B {}_BA_A$ and ${}_BA_A \otimes_A {}_A(A^*)_B = {}_B(A^*)_B$ are algebra objects in the categories ${}_A\mathsf{Mod}_A(\mathcal{C})$ and ${}_B\mathsf{Mod}_B(\mathcal{C})$, respectively.*

Proof. Talk to Tom about the proof for this. ∎

Add citation for Tom's notes.

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